

Reed Instrument Making

Acoustics, Construction and Practice

From the Reed to the Cabinet

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Introduction

This book brings together a series of technical documents that emerged over many years from practical work in reed instrument making. Sie behandeln die Akustik der Stimmzunge und Kammer, die Strömungsmechanik im Spalt, die Kopplung von Zungen und Kammern, die Materialwahl und Konstruktion von Stimmplatten, Bälgen und Gehäusen sowie die musikalischen Themen Stimmung und Tremolo.

These documents are aimed at reed instrument makers, repairers, and everyone who wants to understand why an instrument sounds as it does -- and what can be changed.

About the Author

My first experience with accordions and their repair came in the early 1980s. What I know about building and restoring these instruments was first passed on to me by an experienced instrument maker -- not through formal training or examinations, but by watching, assisting, and many conversations in the workshop.

By profession I trained as a television engineer. Later I taught electronics and communications technology. This technical background has shaped my approach to instrument making: I think in frequencies, impedances, and resonant circuits -- concepts that apply directly to reed acoustics.

From 2000 I began building accordions. Today, after more than 25 years of experience in building and repairing reed instruments, I want to share the insights gained in this document collection.

How These Documents Came About

Each document arose independently and addresses a specific topic. They were not planned as a book from the outset, but are the result of concrete questions that arose in the workshop: Why does a reed respond poorly? How does chamber geometry affect tone? What happens when multiple reeds sound into the same chamber?

I initially wrote down the answers for myself, supported by physical models. My knowledge of electronics and

communications technology proved useful here -- the analogy between electrical resonant circuits and acoustic resonators is direct and fruitful.

The documents do not claim to be academic treatises. Numerical values are often order-of-magnitude estimates based on simplified models and empirical calibration. They aim to provide qualitative understanding -- not to replace measurements on actual instruments.

Document Overview

The collection is organised into the following documents:

Acoustics and Flow

Dok.	Titel
0002	Flow Analysis: 50 Hz Bass Reed -- Version 8
0003	Impedance Comparison: Free-beating Reed vs. Labial Pipe
0004	Frequency Variation of the Reed through Chamber Coupling
0005	Frequency Shift as Response Indicator
0006	Symbol Reference

Chamber and Tone

Dok.	Titel
0007	Treble Reed Block -- Chamber Frequencies D3-C6
0008	Tone Modification by Chamber Geometry
0009	Frequency Coupling of Multiple Reeds

Reed and Reed Plate

Dok.	Titel
0010	Reed Quality Factor
0011	Channel Geometry of the Reed Plate
0012	Reed Stiffness

Coupling and Overtone Modes

Dok.	Titel
0015	Acoustic Coupling and Impedance Matching
0016	Overtone Modes of the Bass Reed: Profiling and Inharmonicity
0017	Treble Reeds: Overtone Modes F3 to C6
0018	Audibility of Bending Modes: Chamber Notch, Transients, Torsion

Tuning and Tremolo

Dok.	Titel
0019	Tuning and Difference Tones
0020	Tremolo: Beat Physics, Types and Tuning Practice

Construction and Materials

Dok.	Titel
0021	Reed Plates: Quality, Manufacturers and Grades
0022	Bellows: Cross-Section, Fold Count and Instrument Size
0023	Cabinet and Mechanism
0024	Practical Notes: Critical Techniques in Instrument Making

Special Topics

Dok.	Titel
0500	Guide to Aesthetic Forensics of Accordion Cases

Notes on Reading

Documents can generally be read in any order. Cross-references at the foot of each document show which other documents address related topics. The symbol reference in Document 0006 summarises the notation used and serves as a reference guide.

For many documents, Python calculation scripts are available on GitHub. These scripts allow the given numerical values to be verified independently and compared with your own measurements.

Johann Pascher — Linz, 2025

Dok. 0002

Flow Analysis: 50 Hz Bass Reed

Version 8

Flow Analysis of a 50 Hz Bass Reed

v8 - v7 base + slot entry loss (K3b) + dynamic gap change (K6)

Corrections relative to v6: (1) gap area $W \cdot h/3$ instead of $W \cdot h/2$ (cantilever profile), (2) laminar friction $f = 64/Re$ instead of $f = 0.03$, (3) threshold pressure $8EIh/(WL^4)$, (4) Helmholtz with slot as neck, (5) Reynolds numbers stated throughout. New in v8: (K3b) slot entry loss $\zeta_{\text{entry}} = 0.5$ added, (K6) static gap narrowing under bellows pressure with critical pressure Δp_{crit} .

1 Construction

In the bass section of an accordion, dozens of reeds sit in individual chambers, all connected to the same bellows. The bellows generates the pressure; the keys control the valves via mechanical levers. The valve is **not** opened by bellows pressure – pressure acts on all closed valves simultaneously. Only pressing a key opens the valve mechanically.

2 Chamber, Reed Plate, and Offset

The chamber is a wooden block measuring $94 \times 48 \times 40$ mm. The entire top surface is sealed by an aluminium reed plate. This plate is **wedge-shaped**: 13 mm thick at the free reed tip (above the valve), only 2 mm at the clamping end. The slot in the plate is therefore 13 mm deep at the free end – air must traverse a short channel before reaching the reed.

The steel reed ($70 \times 8 \times 0.354$ mm) is fastened to the plate with **two screws** (smaller reed plates are riveted). The free end is **offset by 1.5 mm** – at rest it stands 1.5 mm above the plate surface. This creates a triangular gap: 0 mm at the clamp, 1.5 mm at the free tip.

The effective gap area accounts for the cantilever bending profile $y(x) \sim y_{\text{tip}} \cdot (x/L)^2$. The gap is largest at the tip and decreases quadratically:

$$S_{\text{gap}} = \frac{W \cdot h_{\text{max}}}{3} = \frac{8 \times 1.5}{3} = 4 \text{ mm}^2 \quad (1)$$

Stationary & non-stationary analysis

(v6 used factor $1/2 = 6 \text{ mm}^2$ – linear profile.) For comparison: the valve opening area is approx. 244 mm^2 – about 61 times larger. The gap remains the bottleneck.

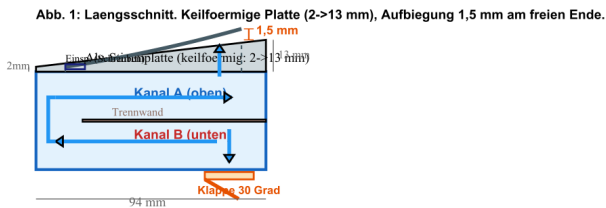


Figure 1: Longitudinal cross-section of the reed. Wedge-shaped plate (2 → 13 mm thickness), 1.5 mm offset at the free tip. The gap (grey) is largest at the tip and decreases quadratically towards the clamp.

3 The Folded Air Path

A dividing wall splits the chamber into two channels. The air path:

Valve (floor, right) → Channel B leftward →
180° fold → Channel A rightward → Slot → Gap

Total path length over 200 mm.

4 Dividing Wall - Film or Veneer?

Two materials are suitable for the dividing wall: **plastic film** (~0.3 mm) or **wood veneer** (~0.5 mm). Wall thickness affects channel width: with a straight wall, film gives 23.85 mm per channel, veneer 23.75 mm – a difference of 0.10 mm (0.4 %). This is negligible from a flow perspective.

Stationary & non-stationary analysis

	Film (0.3 mm)	Veneer (0.5 mm)
Surface	Smooth → less friction ($f \approx 0.02$)	Rough → more friction ($f \approx 0.03$)
Stability	Flexible → can flutter, needs support	Dimensionally stable → holds position
Acoustics	Can vibrate as a membrane → noise	Acoustically inert
Installation	Gluing, tensioning	Gluing, clamping, easier

Table 1: Film vs. veneer as dividing wall material

Since channel friction constitutes a negligible fraction of total losses, the friction difference between film and veneer is irrelevant for flow. The difference lies solely in stability and acoustics. Veneer is the safer choice; film is only appropriate when space is extremely tight.

5 Valve Orientation - Blow-out or Deflect?

Blow-out direction: The valve opens so that the 30° jet points directly into Channel B. The jet has a horizontal component of 87 % and a vertical component of 50 %. Loss coefficient: $\zeta_{\text{valve}} = 0.498$ (vena contracta + direction change).

Deflect direction: The valve opens outwards. Air must turn ~150° around the valve edge before entering the chamber. Loss coefficient: $\zeta_{\text{valve}} = 1.698$ - **3.4 times** the blow-out direction.

	Blow-out	Deflect
Jet	Directed into chamber	First away, then 150° turn
ζ_{valve}	0.498	1.698
Wall utilisation	Full (nozzle, Coanda)	Reduced (wider entry)
Noise	Less	More (separation at edge)
Application	Standard choice	Only when layout requires it

Table 2: Blow-out vs. deflect valve orientation comparison

The higher loss of the deflect direction has little effect on gap velocity (the gap dominates), but reduces flow quality: the jet enters the chamber broader and slower, turbulence at entry is higher, and the wall shape (nozzle at B, Coanda at C) cannot act fully.

Valve Opening and Volume

Practical tests show: reducing the valve opening causes volume losses. The stationary calculation reveals why this is **not** due to flow rate:

Angle	$S_{\text{valve,eff}}$ [mm ²]	$S_{\text{valve}}/S_{\text{gap}}$	ζ_{total}	Q [ml/s]	Q/Q_{30°
30°	244	40.7×	2.0004	912	100.0 %
25°	204	33.9×	2.0005	912	100.0 %
20°	167	27.8×	2.0008	912	100.0 %
15°	126	21.0×	2.0013	911	99.9 %
10°	85	14.1×	2.0032	911	99.9 %
7°	59	9.9×	2.0062	910	99.8 %
5°	43	7.1×	2.0113	910	99.7 %
3°	25	4.3×	2.0297	907	99.5 %
2°	17	2.8×	2.0635	903	99.0 %
1°	9	1.4×	2.2362	885	97.0 %

Table 3: Volume flow at various valve opening angles (at 1000 Pa, wall A)

Stationary flow is remarkably insensitive to valve opening - even at 5° more than 99 % of maximum flow is available. That the practically observed volume losses are **not explainable by stationary flow loss** (the gap dominates too strongly) points to a **non-stationary mechanism**.

1. Slower pressure build-up: A smaller valve opening delays pressure rise in the chamber. The reed takes longer to reach full amplitude. During fast passages the maximum amplitude is never reached - this sounds quieter.

2. Altered jet coupling: At smaller angles the jet is flatter and faster. At 30° the jet has 50 % vertical component, at 10° only 17 %. Less vertical component means the jet is directed less squarely into Channel B.

3. Acoustic impedance change: The valve opening is the acoustic inlet of the chamber. A smaller opening increases acoustic impedance at the chamber inlet, changing the chamber's resonance properties. At certain opening angles, destructive interference can occur that reduces reed amplitude.

All three effects act together and explain why volume losses in practice are more pronounced than stationary calculation would suggest. The calculation shows the effect is not in the flow - so it must be in the non-stationary behaviour.

Acoustic Effect of the 180° Fold

The 180° fold at the end of the dividing wall has sharp 90° corners. Previous analysis considered only the stationary flow loss – which is negligibly small. But the fold has an **acoustic** effect not previously considered.

Coltman Effect: Mitre Bends as Impedance Disturbances

Coltman¹ investigated the acoustic properties of mitre bends in pipes – precisely the geometry used in organ pipes to fold long resonators.

Finding: A sharp 90° mitre bend changes the **characteristic impedance** of the pipe. Reason: the compressibility (volume) of the bend section remains unchanged, but the **inertance decreases** because acoustic oscillations take a “short-cut” around the bend. The result:

- The impedance $Z = \sqrt{\text{inertance}/\text{compressibility}}$ is disturbed
- The effective acoustic length of the section is **shortened**
- A **reflection coefficient** arises at the bend

Coltman measured for a 180° double bend (two closely spaced 90° mitres – exactly the geometry of our chamber fold):

Acoustic shortening: $\Delta l = 18.8 \text{ mm}$ for a pipe with $D_{\text{inner}} = 23.8 \text{ mm}$, thus $\Delta l \approx 0.79 \cdot D$.

Transfer to the Accordion Chamber

The relevant dimension is **not** the hydraulic diameter of the entire fold passage ($D_h = 25.8 \text{ mm}$), but the **narrower dimension at the bend point:** the fold gap $\text{gap}_{\text{fold}} = 19 \text{ mm}$. The acoustic “shortcut” occurs where the wave turns from the channel ($W_k = 23.8 \text{ mm}$ wide) into the fold bow ($\text{gap}_{\text{fold}} = 19 \text{ mm}$) – at the narrower cross-section.

$$\Delta l_{\text{acoustic}} \approx 0.79 \cdot \text{gap}_{\text{fold}} \approx 0.79 \times 19 \text{ mm} \approx 15 \text{ mm}$$

The total internal chamber path (valve → Channel B → fold → Channel A → slot) is approx. 169 mm long. An acoustic shortening of 15 mm equals **9 %** of the acoustic length – not negligible.

¹J. W. Coltman, *Acoustic properties of miter bends*, CCRMA / Stanford (2006). Based on Dequand et al., *Acoustics of 90° sharp bends*, *Acta Acustica* 89 (2003), 1025–1037. Online: <https://ccrma.stanford.edu/marl/Coltman/documents/Coltman-1.44.pdf>

Implications for Corner Geometry

In the old (flow-mechanical) interpretation, the recommendation was: insert a quarter-circle profile to reduce reflections and vortex formation. Acoustically, the situation is more nuanced:

Sharp corners: Maximum inertance reduction, strongest acoustic shortening ($\Delta l \approx 20\text{ mm}$). Highest reflection coefficient. Chamber resonances shift upward.

Rounded corners (quarter-circle R): The impedance transition is smoothed, the reflection coefficient decreases. But simultaneously the inertance reduction is **reduced**, because oscillation paths can “shortcut” less. Acoustic shortening decreases, effective chamber length grows.

Coltman’s compensation: In organ building, the impedance disturbance is not compensated by rounding but by **inserting obstacles**: a thin plate at the dividing wall end (width $\approx 0.5 + 2 \times 8.3 \approx 17\text{ mm}$, height = $H_{\text{chamber}} = 40\text{ mm}$, thickness 2 mm) restores inertance to the value of a straight channel. Alternatively, the sharp corner is chamfered 12 mm at 45° on both sides and covered with a plate ($d \approx 25\text{ mm}$). Both restore the impedance of the straight pipe.

	Sharp	Quarter-circle	Coltman plate
Inertance	Reduced (–)	Partially restored	Fully compensated
Volume	Unchanged	Slightly increased	Reduced (adjusted)
Impedance Z	Disturbed	Differently disturbed	Restored
Reflection	Maximum	Reduced (smoother transition)	Minimal (compensated)
$\Delta l_{\text{acoustic}}$	$\approx 15\text{ mm}$	$\approx 8\text{--}12\text{ mm}$	≈ 0
f_H shift	Upward	Less upward	None

Table 4: Corner geometry comparison – acoustic effect

Comparison with Organ Building

The Organ Historical Society confirms: folded (“mitered”) organ pipes differ acoustically only **slightly** from straight pipes of the same length. The main effect is the acoustic length correction, which is accounted for during voicing. Tone colour (overtone spectrum) changes little – confirming that the impedance disturbance is small at typical ratios ($D/\lambda \ll 1$).

For our chamber at $f = 50\text{ Hz}$: $\lambda = 6.9\text{ m}$, $\text{gap}_{\text{fold}} = 19\text{ mm}$, so $D/\lambda = 0.003$ – the disturbance is small. For overtones ($5f = 250\text{ Hz}$, $\lambda = 1.4\text{ m}$) the ratio rises to 0.014 – still small.

Revised assessment: The fold changes not the flow (stationary loss $< 0.01\%$) but the **acoustic impedance** of the chamber. Sharp corners shorten the effective acoustic length by approx. 15 mm ($\approx 9\%$). A quarter-circle profile reduces reflection but changes impedance **differently** from targeted compensation. In organ building, sharp bends are accepted and compensated via pipe length. The Coltman compensation (horizontal plate $\approx 17 \times 40 \times 2$ mm at the dividing wall end) deliberately restores impedance. In the accordion chamber, where f_H is tuned via chamber volume and neck cross-section, the fold shifts f_H by a few hertz – an effect that should be incorporated into chamber design, not “repaired” after the fact by rounding. Whether rounding improves or worsens response depends on whether the impedance change pushes f_H in the right direction – this can only be determined by experiment.

6 Jet Test - Why Entry Angle Determines Amplitude

Experimental Setup and Observation

An informative practical test: reed plate mounted on chamber, valve removed, blowing with a compressed-air nozzle from various angles and positions. Air always takes the internal chamber path: valve opening $\rightarrow$ Channel B $\rightarrow$ 180° fold $\rightarrow$ Channel A $\rightarrow$ slot $\rightarrow$ gap. Yet reed amplitude changes **from zero to maximum** solely by changing the nozzle angle.

Air does **not** hit the gap directly. The gap sits on the top surface of the reed plate; the flow-relevant underside is accessible only via the chamber interior. **But:** although the jet must take the detour through two channels and a 180° fold, the entry angle at the chamber inlet affects the effective approach flow at the gap – and thus amplitude. Why this is so is shown by the following momentum balance.

Momentum Decomposition at the Chamber Inlet

An air jet entering the valve opening at angle α (measured from horizontal) has two momentum components. At $v_{\text{jet}} = \sqrt{2\Delta p/\rho} \approx 41$ m/s (at 1000 Pa):

Angle α	$\cos \alpha$	v_{horiz} [m/s]	v_{vert} [m/s]	Character
0°	1.00	40.8	0.0	Purely horizontal – maximally directed into Channel B
10°	0.98	40.1	7.1	Shallow – almost all channel flow
20°	0.94	38.3	13.9	Slightly inclined
30°	0.87	35.3	20.4	Normal operation (valve angle)
45°	0.71	28.8	28.8	Symmetric
60°	0.50	20.4	35.3	Steep – much wall impact
90°	0.00	0.0	40.8	Purely vertical – no directed channel flow

Table 5: Momentum decomposition of the entry jet at the chamber inlet

Horizontal component ($v \cdot \cos \alpha$, in Channel-B direction): Becomes directed channel flow. Horizontal momentum is conserved, transported through Channel B, survives the 180° fold (with direction reversal) and arrives as defined flow in Channel A at the gap.

Vertical component ($v \cdot \sin \alpha$, transverse to channel): Is reflected at chamber walls and converted to turbulence. This momentum component dissipates its directional information already in Channel B.

Why Directional Information Survives the Chamber Path
Three reasons:

1. Jet core reaches the fold: The equivalent jet diameter is $d_{\text{jet}} \approx 18 \text{ mm}$ (from $S_{\text{valve}} = 244 \text{ mm}^2$). The core zone, in which the jet retains its direction, extends over $L_{\text{core}} \approx 5d_{\text{jet}} \approx 88 \text{ mm}$. Channel B is only 75 mm long – so the jet core reaches the fold.

2. Laminar channel flow: The Reynolds number in the channel is $\text{Re} \approx 239$ – well below the turbulence threshold. Viscous friction (pressure loss $\Delta p_{\text{friction}} \approx 0.01 \text{ Pa}$, less than 0.001 % of bellows pressure) destroys virtually no directional information.

3. Fold reverses direction but preserves magnitude: Centripetal pressure at the 180° fold is only $\sim 0.02 \text{ Pa}$. The fold reverses horizontal momentum ($+x \rightarrow -x$) but at $\text{Re} \sim 240$ dissipates only a fraction of directed momentum. Estimate: approx. 60 % of directed component survives the fold ($\eta_{\text{fold}} \approx 0.6$).

Effective Approach Flow at the Gap

At the end of Channel A the flow turns $\sim 90^\circ$ into the slot. The **directed portion** of channel flow generates a clear tangential component (parallel to plate, over the gap); the **diffuse portion** distributes evenly. The directional retention factor η_{total} determines how much tangential component arrives at

the gap:

$$\eta_{\text{total}} = \cos \alpha \cdot \eta_{\text{fold}} \cdot \eta_{\text{wall}}, \quad f_{\text{tan}} = \frac{1 + \eta_{\text{total}}}{2} \quad (2)$$

where f_{tan} is the tangential fraction at the gap ($f_{\text{tan}} = 0.5$ for fully diffuse flow, $f_{\text{tan}} \rightarrow 1$ for perfectly directed).

	Wall A (straight)		Wall B (inclined)		Wall C (parabola)	
α	η_{total}	v_{tan} [m/s]	η_{total}	v_{tan} [m/s]	η_{total}	v_{tan} [m/s]
0°	0.180	17.0	0.300	18.7	0.420	20.5
10°	0.177	17.0	0.295	18.7	0.414	20.4
20°	0.169	16.8	0.282	18.5	0.395	20.1
30°	0.156	16.7	0.260	18.2	0.364	19.7
45°	0.127	16.2	0.212	17.5	0.297	18.7
60°	0.090	15.7	0.150	16.6	0.210	17.4
90°	0.000	14.4	0.000	14.4	0.000	14.4

Table 6: Directional retention η_{total} and tangential velocity at gap ($v_{\text{gap}} = 28.8$ m/s). Retention factors: $\eta_{\text{fold}} = 0.6$; η_{wall} : A = 0.3, B = 0.5, C = 0.7.

Bernoulli Drive Power

The Bernoulli suction at the gap - the drive of the oscillation - scales with the square of tangential velocity: $\Delta p_B = \frac{1}{2} \rho v_{\text{tan}}^2$.

	Wall A		Wall B		Wall C	
α	Δp_B [Pa]	Fraction	Δp_B [Pa]	Fraction	Δp_B [Pa]	Fraction
0°	174	34.8 %	211	42.2 %	252	50.4 %
10°	173	34.6 %	210	42.0 %	250	50.0 %
30°	167	33.4 %	198	39.7 %	232	46.5 %
60°	149	29.7 %	165	33.1 %	183	36.6 %
90°	125	25.0 %	125	25.0 %	125	25.0 %

Table 7: Bernoulli suction at gap for various entry angles and wall shapes. Fraction relative to $\Delta p_{B,\text{max}} = \frac{1}{2} \rho v_{\text{gap}}^2 = 499$ Pa (if 100 % tangential).

Numerical values in normal operation ($\alpha = 30^\circ$):

- Wall A: $\eta_{\text{total}} = 0.156$, $v_{\text{tan}} = 16.7$ m/s, $\Delta p_B = 167$ Pa
 - Wall B: $\eta_{\text{total}} = 0.260$, $v_{\text{tan}} = 18.2$ m/s, $\Delta p_B = 198$ Pa
 - Wall C: $\eta_{\text{total}} = 0.364$, $v_{\text{tan}} = 19.7$ m/s, $\Delta p_B = 232$ Pa
- Stationary & non-stationary analysis

Wall C provides **1.4 times** as much Bernoulli drive as Wall A (+1.4dB).

The Flute Mechanism at the Reed Gap

The tangential component flows over the gap opening and generates Bernoulli underpressure that pulls the reed **downward**. Simultaneously the normal component (static pressure in the slot) pushes the reed **upward**.

Crucially: both forces change when the reed oscillates. If the reed moves down (gap widens), more air can flow tangentially over the gap → stronger suction → reed pulled further down. If it moves up (gap narrows), the channel for tangential flow narrows → less suction → reed springs back. This is **positive feedback** that drives the oscillation.

This mechanism is physically related to tone generation in flute instruments: there a flat air jet flows over a sharp edge (labium). The reed edge takes the role of the labium here.

Pressure and Suction Reed: Direction-Independence of the Mechanism

Each reed plate carries **two reeds**: the pressure reed oscillates when air flows from chamber to outside (bellows pressing), the suction reed when air flows from outside into chamber (bellows drawing).

Both oscillate by the same Bernoulli mechanism: gap narrowing generates velocity increase, velocity increase generates underpressure, underpressure pulls the reed into the slot (free-beating reed). The feedback between gap width and underpressure drives oscillation – **independent of global flow direction** through the chamber.

The jet test shows that entry angle affects amplitude. The momentum transport calculation explains a real mechanism: horizontal momentum is transported as directed channel flow to the gap, and wall shape affects retention.

But: The practical finding (Chapter 7) shows that the dividing wall shape acts on pressure *and* suction reeds **equally**. Since flow guidance can only affect the pressure reed (for the suction reed there is no channel jet), the **dominant** mechanism of the dividing wall must be different: the **acoustic impedance coupling** $Z(f)$, which acts identically on both reeds (see Chapter 7). Flow effects are real, but subordinate to acoustic coupling.

7 Pressure and Suction Reed - Why the Dividing Wall Acts Acoustically

Bernoulli at the Constriction: Direction-Independent

Each reed plate carries two reeds for the same pitch. The pressure reed oscillates when pressing the bellows (air from chamber to outside), the suction reed when drawing (air from outside into chamber). Both oscillate by the same Bernoulli mechanism at the constriction:

Pressure reed: $p_{\text{chamber}} > p_{\text{outside}}$, so $\Delta p = p_{\text{chamber}} - p_{\text{outside}}$, and $v_{\text{gap}} = \sqrt{2\Delta p / \rho}$.

Suction reed: $p_{\text{outside}} > p_{\text{chamber}}$, so $\Delta p = p_{\text{outside}} - p_{\text{chamber}}$, and $v_{\text{gap}} = \sqrt{2\Delta p / \rho}$.

At equal bellows pressure $|\Delta p|$ is equal, thus v_{gap} equal, thus Bernoulli suction equal. The reed is a pressure machine, not a momentum machine: the force arises through $F = \Delta p \cdot A$ (pressure field at the constriction), not $F = \dot{m} \Delta v$ (momentum transfer of a jet). Unlike a momentum machine (e.g. Pelton wheel), which fails when suction is reversed, the pressure machine works in both directions because the forced constriction geometry routes all streamlines through the same constriction – regardless of whether driven by pressure or suction.

Practical Finding: Pressure and Suction Behave Identically

In the instrument, response for pressure and suction is **virtually indistinguishable**. This confirms the theory: since the Bernoulli constriction is direction-independent and both reeds use the same gap, identical oscillation behaviour results.

But this has an important consequence for the question of **how** the dividing wall acts:

The Contradiction: Dividing Wall Acts Equally on Both Reeds

The practical test shows: different dividing wall forms affect response – and equally for both pressure **and** suction.

This contradicts the flow-guidance hypothesis from Chapter 6. For if the dividing wall acted via the **directed channel jet** (momentum transport, tangential component, flute effect), it should **only affect the pressure reed**:

- **Pressure reed:** Air flows as jet through valve → Channel B → fold → Channel A → gap. The dividing wall guides this

jet. Directional retention, tangential component at gap – all plausible.

- **Suction reed:** Air is drawn from outside directly into the gap. No jet runs through the channel towards the gap. The dividing wall cannot guide a jet that does not exist.

If the dividing wall nonetheless acts equally on **both** reeds, a different mechanism must dominate.

The Key Test: Unfolded Chamber, 0° vs. 90°

The decisive experimental finding: an **unfolded chamber** (simple tube, no dividing wall, no fold) is placed on the reed plate. The open tube end can be oriented at various angles relative to the vibrating reed tip:

0°: The tube end lies axially, directly at the vibrating reed tip.

90°: The tube end lies transverse to the vibrating reed tip.

Result: Response differs – and for **both** reeds.

In this test there is: no dividing wall, no fold, no directed channel jet, no tangential component, no flute effect. Yet the orientation of the tube end relative to the gap affects oscillation. The only mechanism that can explain this: **the acoustic impedance coupling between chamber opening and gap.**

The Actual Mechanism: Acoustic Impedance Coupling

The chamber acts as a Helmholtz resonator (or at higher frequencies as a $\lambda/4$ resonator). With each oscillation period the reed generates a pressure wave that runs into the chamber, is reflected, and returns. The chamber impedance $Z(f)$ determines:

- The **phase** of the reflected wave relative to reed motion (determines whether energy is fed in or extracted)
- The **amplitude** of the feedback (determines how strong the effect is)

What **pipe position** (0° vs. 90°) changes: where the acoustic “end” of the chamber lies relative to the location of maximum reed deflection affects how the pressure wave couples to the gap. The open pipe end is a pressure node (reflection with phase reversal). Whether this pressure node lies directly at the vibrating reed tip (0°) or laterally offset (90°) changes the effective acoustic length and thus $Z(f)$.

Stationary & non-stationary analysis

What the **dividing wall** changes: it splits the chamber into two channels with different cross-sections. This changes:

- The effective **volume distribution** between channels
- The **Helmholtz frequency** f_H of the overall system
- The **quality factor** Q_H of the resonator (dissipation at walls and edges)
- The **impedance** $Z(f)$, which **both** reeds experience

This mechanism acts on pressure and suction reeds equally, because in both cases acoustic feedback runs through the same chamber with the same impedance.

Summary: The dividing wall affects response **not primarily via flow guidance** (momentum transport, directional retention, tangential component), but via the **acoustic chamber impedance** $Z(f)$. This is proven by the practical finding that dividing wall form acts equally on pressure *and* suction reeds – and by the 0°/90° test, where without dividing wall and without fold, solely the position of the acoustic end relative to the gap changes response. Flow effects (Chapter 6) are real, but subordinate to acoustic coupling.

8 Three Dividing Wall Variants

A - Straight: Both channels 23.8 mm. 30° jet impacts ~90° on wall. Stagnation point, separation, strongest directional loss and reflection.

B - Inclined ($\beta = 3.7^\circ$): Channel B at valve 14.2 mm (nozzle), at fold 23.8 mm. Channel A at gap 33.4 mm (wide). Diffuser half-angle only $3.7^\circ \rightarrow$ safely separation-free.

C - Parabolic: Same endpoints as B, but continuous curvature. Coanda effect guides jet along convex wall. Best directional retention of pressure pulse.

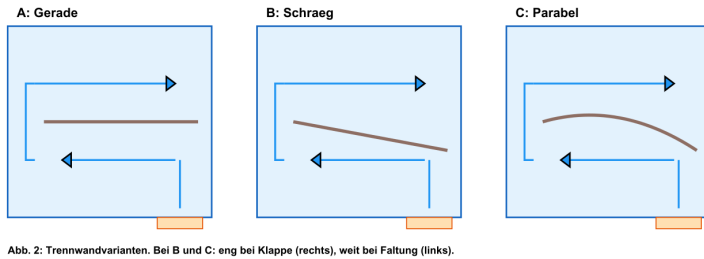


Figure 2: Dividing wall variants. For B and C: narrow at valve (right), wide at fold (left).

All three variants deliver the same volume flow (~ 115 ml/s) and same gap velocity (~ 28.8 m/s) at 1000 Pa. Reynolds numbers: $Re_{\text{gap}} \approx 950$ (laminar), $Re_{\text{channel}} \approx 239$ (laminar, $f = 64/Re = 0.27$). Differences in ζ_{total} are in the per-mille range.

Caution: These results show only the stationary state. The nearly identical volume flows conceal that the dividing wall decisively affects transient response in practice.

Why the Dividing Wall Is Decisive Anyway

The contradiction resolves when two mechanisms are distinguished – one dominant and one subordinate:

Dominant mechanism: acoustic impedance. The dividing wall changes the effective acoustic geometry of the chamber: volume distribution between channels, position of the “acoustic end” relative to the gap, Helmholtz frequency f_H and quality factor Q_H . These changes act via chamber impedance $Z(f)$ on **both** reeds equally (Chapter 7). The $0^\circ/90^\circ$ test with a simple tube – without dividing wall, without fold – shows that solely the position of the chamber opening relative to the gap changes response. The different wall forms (A, B, C) generate different acoustic geometries and thus different $Z(f)$ profiles.

Subordinate mechanism: flow guidance (pressure reed only). In the first milliseconds after valve opening, a pressure pulse runs from the valve through Channel B, around the fold, through Channel A to the gap. The wall form influences how quickly and with what amplitude this pulse arrives at the gap. But this effect acts only on the pressure reed (for the suction reed there is no channel jet) and is subordinate to

acoustic coupling, as the practically identical pressure/suction response shows.

Stationary flow: Identical for all three variants. Therefore flow calculation is useful (it confirms the variants do *not* differ stationary), but it does not explain the audible differences.

The three variants thus differ primarily in their **acoustic effect**:

Straight wall (A): Symmetric channel distribution. Stagnation point generates vortices → increased dissipation → lower Q_H → broader, weaker coupling. Acoustically: uniform impedance profile.

Inclined wall (B): Asymmetric channel distribution (narrow at valve, wide at fold). Less dissipation than A. Changes where acoustic wave encounters different cross-sections → changes $Z(f)$.

Parabolic wall (C): Continuous cross-section change. Minimal reflections within chamber → highest Q_H → narrower, stronger coupling. Acoustically: smoothest impedance transition.

In the electrical engineering analogy: stationary calculation corresponds to the DC resistance of a cable – nearly equal for all three variants. But **impedance** at operating frequency (and its overtones) depends on wave impedance, reflections, and dispersion – which are completely different. For response, DC resistance does not count; the frequency-dependent impedance $Z(f)$ does.

9 Dynamic Gap Area - Why 6 mm^2 Is Only Half the Story

All previous calculations use the rest-position gap area $S_{\text{gap}} = 4 \text{ mm}^2$. But the reed oscillates – and at full volume the reed tip swings approx. 20 mm downward through the slot. In doing so the reed releases the **entire slot cross-section**.

Geometry of the Through-swing Process

The slot has width $W_{\text{slot}} = 9 \text{ mm}$ and runs through the entire wedge-shaped plate (2–13 mm thick). The maximum slot cross-section (if the reed were completely removed) is $W_{\text{slot}} \times t_{\text{mean}} = 9 \times 7.5 = \mathbf{67.5 \text{ mm}^2}$ – eleven times the rest-position gap area.

Three phases during through-swing:

Tip pos.	State	S_{eff} [mm ²]	S/S_{rest}	Released length
+1.5 mm	Above plate	4.5	1.1×	0 mm
+0.5 mm	Above plate	1.5	0.4×	0 mm
0 mm	Gap closes	7.5	1.9×	0 mm
-2 mm	In slot	7.5	1.9×	0 mm
-5 mm	In slot	7.5	1.9×	0 mm
-10 mm	In slot	7.5	1.9×	0 mm
-15 mm	Mostly free	24.7	6.2×	13 mm
-20 mm	Mostly free	42.4	10.6×	28 mm
-25 mm	Mostly free	49.7	12.4×	36 mm
-30 mm	Mostly free	53.7	13.4×	41 mm

Table 8: Effective flow cross-section at various tip positions

Phase 1 - Reed above plate (+1.5 to 0 mm): Normal gap operation. S decreases from 4.5 to 0 mm². Previous calculations apply in this phase.

Phase 2 - Reed in slot (0 to ~-13 mm): Reed plunges into slot. Main cross-section is blocked; only narrow side gaps (0.5 mm each) remain open. S is limited to approx. 7.5 mm² (side gap cross-section). This phase is the **bottleneck in the oscillation cycle**.

Phase 3 - Reed below plate (below -13 mm): Reed tip has left the slot. From the tip, the full slot cross-section is released ($W_{\text{slot}} \times t_{\text{plate}}$). At -20 mm tip: $S_{\text{eff}} = 42 \text{ mm}^2$ (**7-fold**), 28 mm of reed length are free. At -30 mm: $S_{\text{eff}} = 54 \text{ mm}^2$ (**9-fold**).

Time Profile Over the Oscillation Cycle

Ampl.	tip _{min}	<i>S</i> _{min}	<i>S</i> _{max}	<i>S</i> _{mean}	Mean/ <i>S</i> _{rest}
1 mm	−0.5 mm	0.0	7.5	4.3	1.1×
2 mm	−2.5 mm	0.1	7.5	5.6	1.4×
5 mm	−8.5 mm	0.1	7.5	6.4	1.6×
10 mm	−18.5 mm	0.1	38.8	14.9	3.7×
15 mm	−28.5 mm	0.2	52.7	25.4	6.3×
20 mm	−38.5 mm	0.3	57.5	31.5	7.9×
25 mm	−48.5 mm	0.0	59.9	35.4	8.8×

Table 9: Time-averaged gap area over one oscillation cycle (all *S* in mm²)

At full volume (*A* ~ 20 mm), the flow cross-section oscillates between nearly zero and 58 mm² – a **factor of over 100** within a single oscillation period of 20 ms. The time-averaged cross-section is 32 mm² – almost **8 times** the rest-position value (4 mm²).

Consequences for Flow Analysis

1. Gap dominates only during start-up: At the beginning of oscillation (small amplitude) the gap at 4 mm² is genuinely the bottleneck – the chamber with ~950 mm² channel cross-section is irrelevant. But once full amplitude is reached, the gap is wide open for a fraction of each half-period. In this phase the chamber, the valve (244 mm²), and the dividing wall become the relatively narrowest points.

2. Volume flow pulsates massively: During a cycle, flow alternates between virtually zero (reed closes gap) and a maximum far exceeding the stationary value (because *S* is so much larger). Non-stationary volume flow is not a gentle sine signal – it has extremely steep flanks. These pulses must pass through the chamber.

3. Chamber becomes a buffer: The chamber with its volume of approx. 180 cm³ must buffer the extreme volume-flow fluctuations. When the gap briefly allows 10× as much air as in the stationary case, this air must come from somewhere – from the chamber volume. Chamber pressure drops briefly during

each through-swing pulse and is replenished through the valve.

4. Valve area becomes relevant at full volume: In the stationary picture the valve with $S_{\text{valve}}/S_{\text{gap}} = 61\times$ is completely oversized. But when the gap briefly has 58 mm^2 , the ratio is only $S_{\text{valve}}/S_{\text{gap,dyn}} = 4.2\times$. This makes valve opening relevant – and directly explains why a smaller valve opening limits amplitude at full volume (Chapter 5).

5. Dividing wall sees pulsating back-pressure: During each through-swing phase an underpressure pulse arises in the chamber (air is evacuated through the open gap). This pulse runs back through the chamber to the valve. How the chamber transports this pulse (reflections, damping, dispersion) depends on dividing wall form – exactly like the pressure pulse during start-up. The dividing wall form thus affects not only start-up but also **steady-state operation at full volume**.

Stationary calculation with $S_{\text{gap}} = 4\text{ mm}^2$ reliably describes start-up (small amplitude) and gives the correct relations between variants. At full volume the picture changes fundamentally: gap no longer dominates alone, chamber and valve become players, and volume flow becomes extremely pulsating. In this phase chamber geometry becomes relevant even for **stationary** (time-averaged) resistance – not only for the impulse response.

10 Bernoulli Self-Excitation and Start-up Time

Self-excitation occurs via the Bernoulli mechanism: if the gap narrows, v rises, p falls, underpressure amplifies deflection. At the reversal point the flow breaks down, the spring drives the reed back.

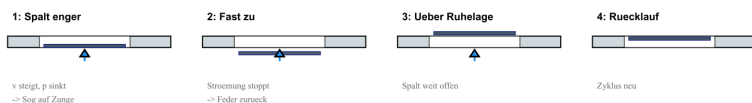


Abb. 3: Bernoulli-Kreislauf (Querschnitt quer zur Zunge)

Figure 3: Bernoulli cycle (cross-section transverse to reed).

Threshold pressure (v7, correct distributed load on cantilever):

$$\Delta P_{\min} = \frac{8EI \cdot h_{\text{offset}}}{W \cdot L^4} \approx 224 \text{ Pa}$$

(3)

Low compared to 500–5000 Pa bellows pressure (v6 had 168 Pa with incorrect lever arm).

Start-up time $\tau = Q/(\pi f)$ is the actual bottleneck:

ζ	Q (quality factor)	τ [ms]	Periods
0.005	100	637	$100 \times 20 \text{ ms}$
0.01	50	318	$50 \times 20 \text{ ms}$
0.02	25	159	$25 \times 20 \text{ ms}$
0.04	12	80	$12 \times 20 \text{ ms}$

Table 10: Start-up time at 50 Hz for various quality factors

Q depends on material, geometry, and clamping and must be measured.

11 Static Gap Narrowing Under Bellows Pressure (v8 new)

Previous calculation uses the rest-position offset $h_{\max} = 1.5 \text{ mm}$ as a fixed value. But **before** the reed begins to oscillate, bellows pressure is already present. This pressure acts as a distributed load on the cantilever and bends the reed tip downward – the gap narrows:

$$w_{\text{tip}}(\Delta p) = \frac{\Delta p \cdot W \cdot L^4}{8EI}, \quad h_{\text{eff}} = h_{\max} - w_{\text{tip}}(\Delta p)$$

(4)

Δp [Pa]	Δp [mbar]	w_{tip} [mm]	h_{eff} [mm]	S_{gap} [mm ²]	S/S_{rest}
100	1.0	0.670	0.830	2.21	0.55
200	2.0	1.340	0.160	0.43	0.11
224	2.2	1.500	0.020	0.05	0.01
500	5.0	3.351	0.020	0.05	0.01
1000	10.0	6.702	0.020	0.05	0.01

Table 11: Static gap narrowing under bellows pressure (K6). Gap closes above $\Delta p > 224 \text{ Pa}$.

Critical pressure: The gap closes at $\Delta p_{\text{crit}} = 8EIh_{\text{max}}/(WL^4) \approx 221 \text{ Pa} \approx 2.2 \text{ mbar}$. This is **identical** to the threshold pressure $\Delta p_{\text{min}} = 224 \text{ Pa}$ – and not by coincidence: both formulae calculate the same physical effect (distributed load that compensates the offset).

In practice, oscillation begins well before Δp_{min} , because Bernoulli instability does not onset only at full static closure:

1. Dynamic Bernoulli effect: Already at approx. 30–50 % of Δp_{min} , flow through the narrowing gap generates enough Bernoulli suction to trigger the first oscillation deflection.

2. Nonlinear amplification: Less gap $\rightarrow$ higher velocity $\rightarrow$ stronger Bernoulli suction $\rightarrow$ further narrowing. This positive feedback loop makes the system unstable before static closure.

3. Estimated onset: Experience values (Fletcher, Cottingham) show that free-beating reeds typically begin oscillating at 30–50 % of static threshold pressure. For this reed: $\Delta p_{\text{dyn}} \approx 90 \text{ Pa} (\approx 1 \text{ mbar})$. At this pressure the gap is still 60 % of rest-position open.

Practical consequence: Static gap narrowing explains why the effective bellows-pressure range of a reed is limited. Below $\sim 90 \text{ Pa}$ the reed does not respond; above $\sim 224 \text{ Pa}$ the gap closes completely. The operating range can be shifted by changing offset: more offset $\rightarrow$ higher Δp_{crit} $\rightarrow$ wider range, but also more air consumption at idle.

12 Damping - What It Affects and How

Quality factor Q determines start-up time and thus response. Q is not a material constant but the sum of all energy losses per oscillation cycle. These can be decomposed into four physically distinct mechanisms:

1. Material damping (internal friction): Every metal converts a fraction of elastic energy to heat when deformed. Loss factor η is a material property. For spring steel η is 0.0002–0.001, for brass 0.001–0.003. In quality factor decomposition: $Q_{\text{material}} = 1/\eta$. Spring steel alone achieves $Q_{\text{material}} \sim 1000$ –5000, brass 300–1000. Frequency-independent.

2. Air damping (viscous losses in gap): The oscillating reed displaces air through the narrow gap (squeeze-film effect). Dissipated power scales with $P_{\text{visc}} \sim \mu WL^3 \omega^2 a^2 / h^3$. With 1.5 mm offset the squeeze-film contribution decreases

Stationary & non-stationary analysis

with h^{-3} , by factor $(1.5/0.4)^3 \approx 53$ compared to a 0.4-mm gap. Subordinate contribution.

3. Clamping damping: At the screw fastening (two screws for bass reeds), energy is microscopically converted to friction during each oscillation. One of the largest individual items. Clean contact surface and uniform screw pressure minimise it.

4. Sound radiation: The oscillating reed radiates sound. At 50 Hz and 70×8 mm area the radiation resistance is low (wavelength 6.9 m). Small contribution for deep bass, increases with f^2 .

Overall quality factor results from harmonic sum: $\frac{1}{Q_{\text{total}}} = \frac{1}{Q_{\text{mat}}} + \frac{1}{Q_{\text{air}}} + \frac{1}{Q_{\text{clamp}}} + \frac{1}{Q_{\text{rad}}}$. For a typical bass reed of spring steel, Q_{total} is likely between 50 and 150.

Q can be **lowered** (faster response) by: material choice (brass instead of steel), tip mass loading, smaller offset, softer clamping. Q can be **raised** (longer sustain, louder tone) by: harder steel, more precise clamping, larger offset. Measurement (decay test) is the only reliable method.

13 Lessons from Propeller and Wind Power Aerodynamics

Trailing edge optimisation: Wind turbine rotors use serrated trailing edges to reduce tonal noise. At a sharp edge, vortices shed more coherently at a single frequency (Kármán vortices). Serrated edges break up this coherence. The same principle applies to the trailing edge of the dividing wall at the fold gap.

Leading edge optimisation: Humpback whale-inspired leading edges (tubercles) prevent simultaneous flow separation across the entire span. Applied to the chamber: a wavy wall edge (instead of the straight edge of Variant A) could distribute the large-scale separation into many smaller, locally limited separations.

Diffuser geometry: For wind energy diffusers the half-angle must remain below 7° . The inclined dividing wall (Variant B) at 3.7° is safely below this. Continuous curvature (Variant C) is superior to a straight diffuser because the pressure gradient is more evenly distributed.

Transferability has limits: propellers operate at $Re = 10^5$ – 10^7 , the reed chamber at $Re \sim 50$ – 1500 . Nevertheless, geometric principles (continuous curvature, avoiding sharp edges, diffuser angle) are valid independent of Reynolds number.

14 Why Resonant Chambers Worsen Response

A common idea is that a resonant chamber could support the reed – similar to how a guitar body functions. Practical measurements show the opposite.

The explanation lies in the type of coupling:

Parallel coupling (L and C in parallel): Both elements are excited simultaneously, energy shuttles. Impedance maximum at resonance. A guitar body works this way.

Series coupling (L and C in series): Energy must pass *through* both elements. At resonance total impedance drops, but each element has its own time constant.

The reed chamber is **series-coupled**. Air must traverse the path bellows → valve → Channel B → fold → Channel A → slot → gap → reed in exactly this sequence. Every chamber resonance adds its own time constant to the system.

What happens with a resonant chamber? When the Helmholtz frequency is a multiple of the reed frequency, the chamber can co-oscillate. Energy is diverted from the main oscillation into the chamber eigenoscillation. For amplitude build-up this acts like additional damping.

Practical consequence: **The chamber should be acoustically as invisible as possible.** Its eigenfrequencies should lie far from the reed frequency and its overtones. With $f_H = 274$ Hz (v7: slot as neck, plate thickness as neck length) for a 50-Hz reed, the ratio is 5.5 – not an integer multiple. This is favourable, but closer to integer multiples than in v6 ($f_H = 461$ Hz).

Practical experience shows that the dividing wall form decisively affects start-up behaviour – although all variants deliver nearly identical stationary volume flows. The explanation lies in series coupling: the chamber determines not flow rate but acoustic impedance at the gap.

Optimisation has two levels: the reed itself (Q , offset, material) determines fundamental response. Chamber geometry (volume, neck cross-section $\rightarrow f_H$) determines the phase of energy supply. Wall form modulates chamber quality factor Q_H . Everything must fit together.

15 Why Phase Determines Response - Not Flow Quality (v8 new)

The previous argument (*clean flow guidance = better response*) is too simplistic. **What matters is not how evenly the flow arrives at the gap, but whether the energy supply arrives at the reed with the correct phase.** Perfectly even flow helps nothing if phase is wrong.

Why no transmission-line model: The gap-to-channel transition ($4 \text{ mm}^2 \rightarrow 950 \text{ mm}^2$) reflects 98.8 % of wave energy – equally for all three variants. A pressure pulse never makes it through this impedance jump as a coherent wave. The chamber does not act as a waveguide but as a **Helmholtz resonator** with impedance $Z_{\text{chamber}}(f)$.

Phase of this impedance determines whether energy is pumped into the reed:

$$\text{Phase}(Z) = \arctan\left(Q_H \cdot \left(\frac{f}{f_H} - \frac{f_H}{f}\right)\right) \quad (5)$$

At $f = f_H$: phase = 0° (resonance, maximum energy exchange). Far below: $\sim -90^\circ$ (compliance). Far above: $\sim +90^\circ$ (mass). For the 50-Hz reed: $f/f_H = 0.18$, phase $\approx -89^\circ$ – almost pure compliance, the chamber does not disturb the fundamental.

Overtone n	f [Hz]	f/f_H	Phase Z [$^\circ$]	Coupling
1	50	0.18	-88.5	Decoupled
5	250	0.91	-52.7	Weak
6	300	1.09	$+51.2$	Weak
8	400	1.46	$+79.5$	Decoupled
20	1000	3.64	$+87.6$	Decoupled

Table 12: Phase of chamber impedance at overtones of a 50-Hz reed ($f_H = 274 \text{ Hz}$, $Q_H \approx 7$)

No overtone of the 50-Hz reed hits $f_H = 274\text{ Hz}$ exactly. The 5th and 6th overtone bracket the resonance with $\sim 53^\circ$ phase offset – noticeable but not resonant.

What wall form really affects: Not phase (determined by f_H and Q_H) but effective chamber quality factor Q_H . Straight wall (A): stagnation point $\rightarrow$ energy dissipation $\rightarrow Q_H$ drops $\rightarrow$ resonance band **wider** (many overtones weakly affected). Parabolic wall (C): less dissipation $\rightarrow Q_H$ higher $\rightarrow$ resonance band **narrower** (few overtones strongly affected). The instrument maker primarily tunes f_H (chamber volume), wall form is fine-tuning of bandwidth.

Why the Fold Changes Phase

The 180° fold is an **acoustic discontinuity**. At the sharp deflection the effective cross-section changes abruptly and flow direction reverses. A pressure pulse is partially reflected there. The reflected portion runs back to the gap and superimposes on the direct chamber pressure.

This superposition is the mechanism: the reflected pulse arrives with a time delay (double fold distance / c). Depending on frequency the superposition is constructive or destructive – the **effective phase** of chamber pressure at the gap shifts. Additionally, the fold changes the effective acoustic mass of the neck (the air path becomes longer and more convoluted), which slightly shifts f_H .

Rounding (quarter-circle) reduces the reflection coefficient at the fold but simultaneously changes the inertance of the bend section (Coltman effect, Section 5.2). A continuous cross-section transition generates less reflection than an abrupt kink – but also less acoustic shortening. This means: phase comes closer to the ideal Helmholtz value (more predictable) but f_H shifts downward (effective chamber length grows). Whether this improves response depends on where f_H lies relative to reed overtones.

Hierarchy of Effects for Response

Chamber geometry affects response via **several mechanisms**, physically distinct and of different importance:

1. Acoustic phase coupling (dominant): Phase of chamber impedance $Z(f)$ determines whether reed overtones absorb or donate energy to chamber. Determined primarily by

f_H (chamber volume, neck cross-section, neck length). Acts on **both** reeds (pressure and suction) equally. This is the main reason **chamber size** varies for different pitches.

2. Acoustic geometry of dividing wall (secondary, but practically relevant): Dividing wall changes effective acoustic chamber geometry – where the “acoustic end” lies relative to gap, how volume distribution between channels is arranged, and how quality factor Q_H of the resonator turns out. Higher Q_H (less dissipation, e.g. Variant C) → narrower, stronger coupling. Lower Q_H (more dissipation, e.g. Variant A) → broader, weaker coupling. That this effect acts equally on **both** reeds confirms the acoustic mechanism (Chapter 7). The 0°/90° test with unfolded chamber shows that solely position of the chamber end relative to gap changes response – without dividing wall, without fold.

3. Flow guidance (subordinate): Momentum transport through the channel (Chapter 6: tangential component, directional retention) is a real effect, but affects only the pressure reed. Since pressure and suction response are practically indistinguishable and dividing wall form acts equally on both, this mechanism is subordinate to acoustic coupling.

The **practical experience** of instrument makers that dividing wall form acts audibly is explained primarily by the acoustic mechanism (points 1 and 2): wall form changes $Z(f)$ and Q_H , and this change acts on both reeds. Flow effects (point 3) are a real but quantitatively small addition affecting only the pressure reed. For optimisation: chamber geometry (volume, neck) is more important than dividing wall form, and acoustic effect of the dividing wall is more important than its flow-technical effect.

16 Pre-Chamber Between Valve and Main Chamber (v8 new)

What happens when an additional chamber ($48 \times 20 \times 70$ mm, $V_{PC} = 67.2 \text{ cm}^3$) is built between valve and main chamber? Coupling occurs via an opening with the dimensions of the valve opening ($S = 244 \text{ mm}^2$).

Coupled Helmholtz Resonators

Two chambers with connecting neck form a coupled two-mass system. The eigenfrequencies split:

	Without PC	With PC
Mode 1 (in-phase)	274 Hz	231 Hz
Mode 2 (anti-phase)	–	895 Hz
Anti-resonance	–	~753 Hz

Table 13: Eigenfrequencies without and with pre-chamber (PC)

Mode 1: Both chambers oscillate in-phase. Reduced 16 % from original (274 Hz) – the additional volume increases effective compliance. **Mode 2:** Chambers oscillate anti-phase at 895 Hz – far above the relevant range.

Phase Shift at the Gap

OT n	f [Hz]	Phase w/o PC	Phase w/ PC	Change
1	50	-88.5°	-87.7°	~same
5	250	-52.7°	$+41.7^\circ$	better
6	300	$+51.2^\circ$	$+71.3^\circ$	worse
8	400	$+79.5^\circ$	$+81.0^\circ$	~same

Table 14: Phase of chamber impedance at slot, with and without pre-chamber ($Q_H \approx 7$)

The PC shifts Mode 1 from 274 to 231 Hz. The 5th overtone (250 Hz), which without PC lay just below resonance, now sits just above – phase jumps from -53° to $+42^\circ$ (closer to $0^\circ \rightarrow$ better coupling). The 6th overtone (300 Hz) moves further from resonance: $+51^\circ \rightarrow +71^\circ$ (worse coupling).

Practical recommendation: The pre-chamber is a **tuning tool**. By varying PC length, Mode 1 can be shifted continuously. Since the PC does not affect stationary flow (ζ_{coupling} relative to gap: 4×10^{-4}), it is pure acoustic tuning without side effects on fundamental response. Optimum: choose PC length so that an important overtone falls on Mode 1 (phase = 0°).

17 Which Bass Tones Are Disadvantaged by Chamber Resonance?

The Helmholtz frequency of this chamber is 274 Hz. Which bass tones have an overtone falling near this resonance? As explained in Chapter 10, a resonant chamber extracts energy from the reed via series coupling.

Bandwidth of the Helmholtz resonance depends on its own quality factor Q_H . A wooden chamber with valve typically has $Q_H \sim 5\text{--}10$. At $Q_H = 10$ the critical band is 274 ± 23 Hz, at $Q_H = 5$ it widens to ± 27 Hz.

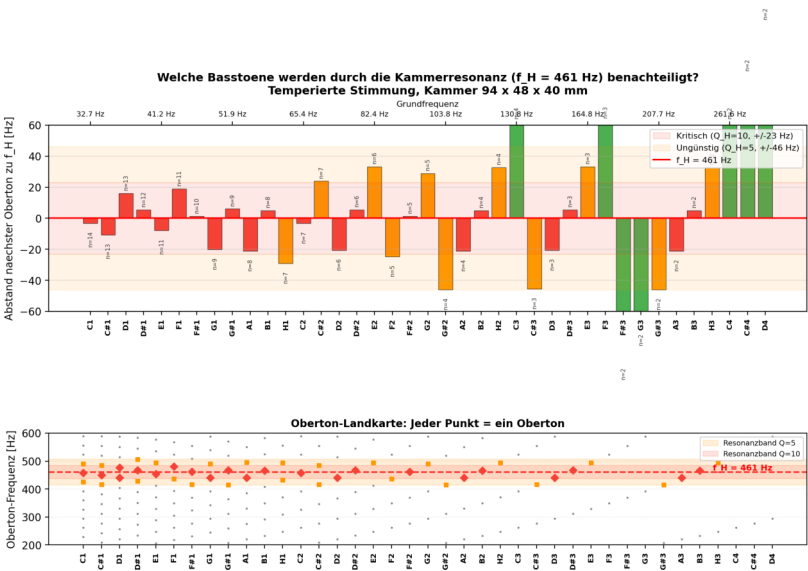


Figure 4: Upper diagram: distance of nearest overtone to f_H . Red = critical, orange = unfavourable, green = unaffected. Lower diagram: overtone map – each point is an overtone, red diamonds fall within the resonance band.

Below ~ 70 Hz (C2) almost every tone is critically affected.

The reason: at low pitches, overtones lie close together (spacing = fundamental frequency). A 40 Hz tone has overtones at 40, 80, 120, ..., 440, 460, 480 Hz – the grid is so fine that an overtone always falls within the resonance band. The lower the pitch, the more unavoidable the conflict.

Particularly affected are tones whose fundamental frequency is an integer divisor of f_H :

Stationary & non-stationary analysis

f_H/n	f [Hz]	Nearest note	Difference	Overtone n
274/2	137.0	C3 (130.8)	+6.2 Hz	2
274/3	91.3	F2 (87.3)	+4.0 Hz	3
274/4	68.5	C2 (65.4)	+3.1 Hz	4
274/5	54.8	A1 (55.0)	−0.2 Hz	5
274/6	45.7	F1 (43.6)	+2.0 Hz	6
274/7	39.1	E1 (41.2)	−2.1 Hz	7
274/8	34.2	C1 (32.7)	+1.5 Hz	8
274/9	30.4	C1 (32.7)	−2.3 Hz	9

Table 15: Integer divisors of f_H and corresponding bass tones

Above C3 (130.8 Hz) no divisor lies close enough. But below – especially A1 (divisor 5, deviation only −0.2 Hz!) and F1 (divisor 6) – there are critical tones. At $Q_H = 5$ the resonance band is 274 ± 27 Hz wide, i.e. 247–301 Hz. The lowest bass tones have overtones so closely spaced that an overtone almost always falls within the band.

18 What Calculations Can and Cannot Achieve

The presented calculations are based on well-established physical models: Euler-Bernoulli beam theory for reed frequency, Bernoulli equation for gap flow, Helmholtz resonator for chamber acoustics, loss coefficients from fluid mechanics (Idelchik). These models give results of the right order of magnitude.

Nevertheless they do not replace practical testing:

1. Idealised geometry: Exactly plane surfaces, sharp edges, uniform gaps. In reality: wood grain, machining marks, individual offsets. Even 0.1 mm difference in offset changes S_{eff} by $\sim 7\%$.

2. Stationary flow: This is the most serious limitation. Loss calculation assumes stationary flow. In reality the flow pulsates at 50 Hz. Practical experience shows that dividing wall form decisively affects start-up behaviour. The dominant mechanism is **acoustic** (impedance coupling $Z(f)$, acts on both reeds equally, see Chapter 7), not fluid-mechanical. Stationary calculation captures neither acoustic nor transient effects.

3. Damping: Q is the most sensitive parameter. Differences of factor 2 between apparently identical reeds are nor-

mal.

4. Chamber acoustics: The Helmholtz model is simplified. In reality the chamber has further modes, and acoustic properties change with bellows pressure, temperature, and humidity.

5. Interactions: Every effect is modelled individually. A complete solution would require coupled fluid-structure-acoustics simulation (FSI).

Calculations reliably identify which parameters matter and at which level they act. They rank the three dividing wall variants in a physically justified order (C better than B better than A – confirmed by practical experience). Where calculation reaches its limits is the quantitative prediction of the wall effect: that the wall acts is physically justifiable. How strongly – can only be measured.

For the instrument maker this means: **Calculation says where to look. Practical testing says what to find.** Together, both are more than either alone.

19 Summary

- **Reed plate:** Wedge-shaped (2→13 mm). At the free end, the 13-mm slot acts as a short channel with its own flow resistance ($\zeta \approx 1.62$, including $\zeta_{\text{entry}} = 0.5$ entry loss [K3b]).
- **Gap:** Triangular (0→1.5 mm), $S_{\text{eff}} = W \cdot h/3 = 4 \text{ mm}^2$. Dominates total resistance. All variants: $v \approx 29 \text{ m/s}$, $Q \approx 115 \text{ ml/s}$ at 1 kPa.
- **Dynamic gap change [K6]:** Bellows pressure bends reed statically downward. Critical pressure $\Delta p_{\text{crit}} = 221 \text{ Pa} \equiv$ threshold pressure. Effective oscillation onset at $\sim 90 \text{ Pa}$ (1 mbar). Operating range 90–224 Pa.
- **Wall material:** Film and veneer are equivalent from a flow perspective. Veneer is more stable and acoustically more neutral.
- **Valve orientation:** Blow-out ($\zeta = 0.66$) better than deflect ($\zeta = 1.86$). Blow-out fully utilises wall geometry.
- **Dividing wall form:** Practically decisive for response (non-stationary start-up). C (parabola) best directional retention, B (inclined) good compromise, A (straight) simplest but most sluggish. Stationary flow equal for all – the difference lies in impulse response (reflections, directional loss). Turbulence

for self-excitation is generated by the gap itself.

- **Response:** $\tau = Q/(\pi f)$. At $Q = 100$: 637 ms, at $Q = 25$: 159 ms. Chamber geometry does not affect stationary flow but decisively affects non-stationary start-up.
- **Acoustics:** Helmholtz 274 Hz, chamber acoustically small.
- **Phase coupling (v8 new):** Response by three mechanisms: (1) acoustic phase (dominant, via f_H), (2) acoustic geometry of dividing wall (Q_H , volume distribution, position of acoustic end – acts on both reeds), (3) flow guidance (sub-ordinate, pressure reed only).
- **Pre-chamber (v8 new):** $48 \times 20 \times 70$ mm (67 cm^3) lowers Mode 1 from 274 to 231 Hz (–16%). PC length is tuning tool for overtone phase without effect on stationary flow.

Dok. 0003

Impedance Comparison

Free-beating Reed vs. Labial Pipe

Impedance Comparison

Free-beating Reed vs. Labial Pipe

Supplementary chapter to Doc.0002 (bass_50hz_v8). Systematically presents the fundamental impedance difference between a fluted organ pipe and a reed in a bass chamber. Numerical values refer to the 50 Hz bass reed from Doc. 0002. The resulting frequency variation is calculated quantitatively in Doc. 0004.

Document References

Doc.0001	Analysis of instrument acoustics and psychoacoustics
Doc.0002	Flow analysis of bass reed 50 Hz - v8
Doc.0003	Impedance comparison: free-beating reed vs. labial pipe (this document)
Doc.0004	Frequency variation through chamber coupling - two-filter model
Doc.0006	Symbol reference
Doc.0007	Treble reed block - chamber frequencies
Doc.0008	Tone modification by chamber geometry
Doc.0500	Guide to aesthetic forensics of accordion casework
berechnungen_verifikation.py	Verification script for all calculations

1 Why This Comparison?

The bass chamber of an accordion and a folded organ pipe have much in common externally: a folded air path, a resonance volume, a 180° fold (mitre bend), an opening to the outside air. The Coltman effect (impedance disturbance at the fold, Doc.0002 Section 5.2) applies to both systems. The acoustic shortening $\Delta l \approx 0.79 \cdot D$ is geometric, independent of the excitation mechanism.

Cross-references: Doc. 0002 · Doc. 0004

But the **nature of sound generation** is fundamentally different — and with it, the role that the chamber impedance plays in the oscillation behaviour. This difference explains why organ builders optimise their pipes as resonators, while accordion builders want their chambers to be acoustically as “invisible” as possible (Doc.0002 Ch. 10).

Nevertheless, practical tests clearly show that the chamber measurably shifts the pitch of the reed (1–5 cents). This effect is calculated quantitatively in Doc.0004 as a coupled two-filter system.

2 The Labial Pipe: The Tube Determines the Frequency

Excitation Mechanism

In a fluted (labial) organ pipe, a flat air jet from the core channel creates an instability at the labium (cutting edge). The jet oscillates laterally — driven by the **acoustic feedback from the tube**. The frequency of the tone is **not** determined by the jet (which has no natural frequency in the musical sense), but by the tube resonance.

Input Impedance of the Tube

The tube acts as an acoustic resonator with frequency-dependent input impedance:

$$Z_{\text{tube}}(f) = j \rho c \cdot \frac{S_{\text{mouth}}}{S_{\text{tube}}} \cdot \tan\left(\frac{2\pi f L_{\text{eff}}}{c}\right) \quad (1)$$

(stopped pipe; open pipe: cot instead of tan). The impedance has **sharp maxima** at the tube resonances ($\lambda/4$ for stopped, $\lambda/2$ for open).

Coupling: Jet as Velocity Generator

The jet at the labium is a **velocity generator** (high velocity, low impedance). It “sees” the tube as a load. Oscillation occurs where the **tube impedance is maximal** — because there the greatest pressure feedback acts on the jet. The jet adapts to the tube length.

Key statement - labial pipe: The tube is the frequency-determining resonator. The jet is a broadband energy source that couples maximally at the tube resonance. Impedance at the labium is **constant in time** (linear coupling) — the tube end does not move.

3 The Free-beating Reed: The Reed Determines the Frequency

Excitation Mechanism

The reed is a **mechanical oscillator** with its own natural frequency (Euler-Bernoulli). The 50 Hz bass reed in Doc.0002 has:

$$f_{\text{reed}} = \frac{1.875^2}{2\pi L^2} \sqrt{\frac{EI}{\rho_s A}} \approx 50 \text{ Hz} \quad (2)$$

The frequency is determined by the length, stiffness and mass of the reed — **not** by the chamber. The chamber has $f_H = 274 \text{ Hz}$ — far from 50 Hz. **But** the chamber still shifts the frequency measurably (see Doc.0004, Ch. 3).

The Gap as Nonlinear Impedance Converter

Here lies the fundamental difference. The gap between reed and plate converts the mechanical impedance of the reed into an **acoustic impedance** that the chamber “sees”:

$$Z_{\text{acoust,gap}}(t) = \frac{\Delta p(t)}{q(t)} = \frac{\rho v_{\text{gap}}(t)}{2 S_{\text{gap}}(t)} \quad (3)$$

Since $S_{\text{gap}}(t)$ changes by a **factor of 100** during the oscillation cycle (Doc.0002 Ch. 8: from ~ 0 to 58 mm^2), the acoustic impedance at the gap is **time-variant and strongly nonlinear**.

Mechanical Impedance of the Reed

The reed itself has the mechanical impedance of a damped spring-mass system:

$$Z_{\text{mech}}(f) = R_{\text{damping}} + j \left(\omega m_{\text{eff}} - \frac{k_{\text{eff}}}{\omega} \right) \quad (4)$$

At $f = f_{\text{reed}}$: $Z_{\text{mech}} = R_{\text{damping}}$ (purely real, minimum). The quality factor $Q = \omega_0 m_{\text{eff}} / R$ determines the bandwidth — and thus how strongly the chamber impedance can shift the frequency (Doc.0004 Ch. 2).

Key statement - free-beating reed: The reed is the frequency-determining oscillator. The chamber is **not** a resonator in the sense of frequency selection, but an impedance modulator that influences — through the phase of the feedback — the overtones, the onset time **and the pitch**. Impedance at the gap is **time-variant** (nonlinear coupling). The quantitative calculation of the frequency shift as a coupled two-filter system is found in Doc.0004.

4 Comparison of Impedance Properties

Property	Labial pipe (organ)	Free-beating reed (accor-dion)
Frequency determination	Tube resonance (impedance maximum)	Mechanical natural frequency of the reed
Excitation	Jet at labium (velocity genera-tor, broadband)	Bernoulli suction at gap (self-excited at f_{reed})
Impedance at excitation point	Constant in time (tube mouth does not move)	Periodically switched: factor ~100 per cycle
Coupling type	Quasi-linear (small jet deflec-tion)	Strongly nonlinear ($S_{gap} \sim 0 \dots 58 \text{ mm}^2$)
Chamber/tube role	Frequency-selective (the tube <i>is</i> the instrument)	Phase coupling + frequency pulling (Doc.0004)
Coupling (electronics analogy)	Parallel resonance (LC parallel, jet tube)	Series coupling (air must pass <i>through</i> chamber, Doc.0002 Ch. 10)
Resonant chamber	Desired (resonance = sound production)	Harmful (resonance draws en-ergy, Doc.0002 Ch. 10)
Overtone structure	Forced by tube (harmonic through tube length)	Generated by reed, <i>filtered</i> by chamber
Frequency shift	Unlimited (tube <i>is</i> frequency-determining)	1-5 cents (Doc.0004: $\Delta k = A_{eff}^2 \rho c^2 / V$)
Coltman effect	Correction of tube length ($\Delta l \rightarrow$ intonation)	Correction of f_H ($\Delta l \rightarrow$ phase shift of overtones)

Table 1: Impedance comparison: labial pipe vs. free-beating reed

5 The Gap as Periodically Switched Impedance
Electronics Analogy

In the electronics analogy (Doc.0002 Ch. 7), the labial pipe is a **linear LC oscillator** excited by a constant current source (jet). The resonant frequency $f_0 = 1/(2\pi\sqrt{LC})$ is time-independent.

The free-beating reed, by contrast, is an LC circuit whose **terminating resistance $R(t)$ is switched between zero and infinity at the reed frequency:**

$$R_{gap}(t) = \begin{cases} \infty & \text{gap closed (reed in slot)} \\ \frac{\rho v}{2 S_{gap}(t)} & \text{gap open} \end{cases} \tag{5}$$

This is a **parametric oscillator**: a system in which a circuit parameter (R) is periodically modulated. Doc.0004 shows that it is precisely this parametric modulation that forms the basis of the closed transfer function $H_{\text{ges}}(f) = H_1(f)/(1 - \kappa_{\text{eff}} \cdot H_1 \cdot H_2)$.

Numerical Values for the 50 Hz Bass Reed

Phase in cycle	S_{gap} [mm ²]	v_{gap} [m/s]	Z_{acoust} [Pa s/m ³]
Gap closed	≈ 0	–	$\rightarrow \infty$
Gap minimal (side gap)	7.5	22	1.8×10^6
Gap at rest position	4.0	29	4.3×10^6
Gap maximal (full swing)	58	2	2.1×10^4

Table 2: Acoustic gap impedance at various phases of the oscillation cycle (at 1 kPa bellows pressure). Factor between minimum and maximum: ~ 200 .

The impedance at the gap changes by more than **two orders of magnitude** within a single oscillation period of 20 ms. For a labial pipe, the corresponding variation is $< 1\%$. This is the fundamental physical reason why linear acoustic models (Helmholtz, transmission line) are reliable for the reed only in the small-amplitude regime — and yet a measurable frequency shift arises (Doc.0004).

6 Consequence: Why the Chamber Must Be Optimised Differently

Organ Pipe: Resonance Desired

The organ builder **maximises** the resonance quality of the tube. Higher Q means sharper frequency selection, stronger feedback, less energy loss. Hence organ pipes are made from smooth material, and Coltman compensation serves **exact intonation**.

Reed Chamber: Resonance Harmful — But Not Without Effect

The accordion builder wants the chamber to **not resonate** (Doc.0002 Ch.10). But the chamber is never **without effect**: even when $f_{\text{reed}} \ll f_H$, the chamber impedance acts as a **frequency puller**. Three mechanisms (Doc.0004 Ch.3): (1) Static compliance load (air spring, $\Delta k = A_{\text{eff}}^2 \rho c^2 / V$, +0.3 cents). (2) Resonant amplification at overtones near f_H

Doc. 0003 — Impedance Comparison: Free-beating Reed vs. Labial Pipe
(+1-5 cents at the overtone). (3) Nonlinear feedback to the fundamental (+1-3 cents).

Parameter	Organ pipe (C2, stopped)	Bass reed (50 Hz)
Effective tube length	~1300 mm (stopped $\lambda/4$)	169 mm (chamber path)
$f_{\text{resonance}}/f_{\text{tone}}$	= 1.00 (resonance is the tone)	= 5.48 (f_H/f_{reed} , far away)
Resonance quality Q	20-50 (desired: high)	5-10 (desired: low)
Impedance at mouth	~constant (linear)	factor ~200 per cycle
Frequency shift by resonator	100 % (resonator = pitch)	1-5 cents (Doc. 0004)

Table 3: Quantitative comparison: organ pipe (C2 stopped) vs. bass reed (50 Hz)

7 The Three Time Scales of the Impedance

For the free-beating reed, **three time scales** are relevant that play no role in the labial pipe:

1. Fast time scale — gap modulation ($T = 20$ ms): The impedance at the gap is switched 50 times per second between near zero and near infinity. This effect produces nonlinear self-excitation and determines the overtone spectrum.

2. Medium time scale — transient onset ($\tau \sim 100$ -600 ms): During amplitude build-up, the time-averaged gap area grows (Doc. 0002 Ch. 8: from 4 mm² to 32 mm²). In this phase the Helmholtz approximation is most valid.

3. Slow time scale — bellows pressure (~ 1 s): Bellows pressure statically deflects the reed (Doc. 0002 Ch. 9), shifts the operating point and thus the effective coupling.

8 Why Linear Models Are Still Useful

Despite the strong nonlinearity, the linear Helmholtz model is **not wrong**. It describes the time-averaged impedance and is valid for: transient behaviour, overtone analysis, comparison of partition variants A/B/C (Doc. 0002 Ch. 8), and the calculation of frequency shift in the two-filter model (Doc. 0004).

Where the linear model fails: At full amplitude ($A \sim 20$ mm) the time-averaged gap area becomes ~ 8 times larger. At this stage, a time-resolved FSI simulation would be needed.

9 Summary

Core statement: The reed converts the chamber impedance from a **frequency-selecting resonator** (organ pipe: tube determines frequency, impedance constant in time, resonance desired) into a **phase-modulating energy buffer** (accordion: reed determines frequency, impedance time-variant by factor 200, resonance harmful).

The gap as nonlinear impedance converter is the decisive element with no organ pipe equivalent. **Nevertheless** the chamber measurably shifts the reed frequency (1-5 cents), which can be calculated as a coupled two-filter system (Doc.0004):

$$H_{\text{ges}}(f) = H_1(f)/(1 - \kappa_{\text{eff}} \cdot H_1 \cdot H_2).$$

For the instrument builder: **The chamber is not a resonating body**, but an **impedance environment** that the reed “sees” and that co-determines its frequency. Optimisation targets phase compatibility (Doc.0002 Ch. 11-12) and minimal frequency pulling (Doc.0004 Ch. 5).

The calculation tells you where to look. The practical test tells you what you find.

Dok. 0004

Frequency Variation of the Reed

through Chamber Coupling

Reed Frequency Variation

through Chamber Coupling – Two-Filter Model

Shows why the chamber shifts the reed pitch despite separate natural frequencies, and how this can be calculated as a filter combination of two coupled systems. Numerical values: 50 Hz bass reed, $f_H = 274 \text{ Hz}$, $Q_H \approx 7$ (from Doc.0002).

Document References

Doc. 0001	Analysis of instrument acoustics and psychoacoustics
Doc. 0002	Flow analysis of bass reed 50 Hz – v8
Doc. 0003	Impedance comparison: free-beating reed vs. labial pipe
Doc. 0004	Frequency variation – two-filter model (this document)
Doc. 0006	Symbol reference
Doc. 0007	Treble reed block – chamber frequencies
Doc. 0008	Tone modification by chamber geometry
Doc. 0500	Guide to aesthetic forensics of accordion casework
berechnungen_verifikation.py	Verification script for all calculations

1 The Practical Finding

Every experienced accordion builder knows the phenomenon: **the same reed, mounted in different chambers, sounds at different pitches.** The frequency variation is small (a few cents) but clearly measurable. Cottingham and Millot measured shifts of 10–30 cents in free-beating reeds when the resonator was changed. In accordion bass chambers, where $f_{\text{reed}} \ll f_H$, the effect is smaller (1–5 cents) but systematic.

This contradicts the simplified statement from Doc.0002 Ch.11 that the chamber “does not disturb the fundamental”

Cross-references: Doc. 0002 · Doc. 0003

(phase $\approx -89^\circ$). The phase is indeed almost -90° — but **almost** is not **exactly**, and even a pure compliance shifts the frequency because it imposes an additional spring stiffness on the reed.

Doc. 0003 shows that the fundamental difference from the organ pipe lies in the **time-variant gap** (impedance factor 200 per cycle). Nevertheless, the frequency shift is calculable — with the following two-filter model.

Practical consequence: An instrument is not tuned “on the bench” and then installed in any chamber. The reed is tuned **in its chamber**. If the chamber is changed, the pitch changes.

2 The Two-Filter Model

Two Resonators, One Gap

Filter 1 - The reed (mechanical band-pass): Spring-mass system with $f_1 = 50$ Hz and $Q_1 = 50$ –150 (Doc. 0002 Ch. 9):

$$H_1(f) = \frac{f_1^2}{f_1^2 - f^2 + jff_1/Q_1} \quad (1)$$

Filter 2 - The chamber (acoustic band-pass): Helmholtz resonator with $f_H = 274$ Hz and $Q_H = 5$ –10 (Doc. 0002 Ch. 11):

$$H_2(f) = \frac{f_H^2}{f_H^2 - f^2 + jff_H/Q_H} \quad (2)$$

The coupling: The gap converts reed deflection into volume flow ($q = A_{\text{eff}} \cdot \dot{x}$) and chamber pressure into force ($F = A_{\text{eff}} \cdot p$). The effective gap area $A_{\text{eff}} = W \cdot h_{\text{max}}/3 = 4 \text{ mm}^2$ (Doc. 0002 Ch. 2) is the coupling element.

The Feedback Loop (Barkhausen Condition)

The system is a **feedback loop**:

$$x \rightarrow S(x) \rightarrow q \rightarrow p_{\text{chamber}} \rightarrow F_{\text{Bernoulli}} \rightarrow x$$

The oscillation frequency satisfies the Barkhausen condition:

$$\varphi_{\text{reed}}(f) + \varphi_{\text{chamber}}(f) + \varphi_{\text{coupling}}(f) = 0 \quad (3)$$

Without chamber: $\varphi_{\text{reed}}(f_{\text{mech}}) = 0 \rightarrow$ oscillation at f_{mech} .
With chamber: $\varphi_{\text{chamber}}(f_{\text{mech}}) \neq 0 \rightarrow$ the reed shifts its frequency.

Core statement: The chamber adds a phase shift to the feedback. To satisfy the phase condition, the system shifts its oscillation frequency. The shift depends on how steeply the reed phase varies at f_{mech} ($= Q_1$) and how large the chamber phase is at f_{mech} ($= f_{\text{mech}}/f_H$ and coupling strength).

3 The Three Mechanisms of Frequency Shift

Mechanism 1: Static Compliance Load (Air Spring)

The chamber ($V = 180 \text{ cm}^3$) acts as an additional spring:

$$\Delta k = \frac{A_{\text{eff}}^2 \cdot \rho c^2}{V} = \frac{(4 \times 10^{-6})^2 \times 1.2 \times 343^2}{1.8 \times 10^{-4}} = 0.013 \text{ N/m} \quad (4)$$

Compared with the mechanical spring stiffness $k_{\text{mech}} = \omega_1^2 m_{\text{eff}} = 37.0 \text{ N/m}$ (with $m_{\text{eff}} = 0.243 \times m_{\text{total}} = 0.375 \text{ g}$ for the first cantilever mode):

$$\frac{\Delta f}{f} = \frac{\Delta k}{2k_{\text{mech}}} = \frac{0.013}{74.0} = 1.7 \times 10^{-4} \Rightarrow \mathbf{0.3 \text{ cents}} \quad (5)$$

Always **upward** (stiffer spring), always present, independent of f_H .

Mechanism 2: Resonant Amplification at Overtones near f_H

The impedance is amplified by factor $G(f)$ near f_H :

$$G(f) = \frac{1}{\sqrt{(1 - (f/f_H)^2)^2 + (f/(f_H \cdot Q_H))^2}} \quad (6)$$

OT n	f [Hz]	f/f_H	$G(f)$	Phase Z [°]
1	50	0.18	1.0	−88.5
3	150	0.55	1.4	−83.6
5	250	0.91	4.7	−52.1
f_H	274	1.00	7.0	0.0
6	300	1.09	4.0	+51.8
8	400	1.46	0.9	+79.6
10	500	1.82	0.4	+83.6

Table 1: Impedance amplification $G(f)$ of the Helmholtz resonator at the overtones of the 50 Hz reed ($f_H = 274$ Hz, $Q_H = 7$). Phase values from Doc. 0002 Ch. 11.

Mechanism 3: Nonlinear Feedback to the Fundamental

The overtones are bound to the fundamental through nonlinear gap coupling (Doc. 0002 Ch. 8, Doc. 0003 Ch. 5). If the n -th overtone is pulled by Δf_n :

$$\Delta f_1 \approx \frac{1}{n} \cdot \Delta f_n \quad (7)$$

The Bernoulli Amplification Factor

The coupling constant:

$$\kappa = \frac{A_{\text{eff}}}{\sqrt{m_{\text{eff}} \cdot C_a \cdot \omega_1 \cdot \omega_2}} = \frac{4 \times 10^{-6}}{\sqrt{3.75 \times 10^{-4} \cdot 1.28 \times 10^{-9} \cdot 314 \cdot 1721}} \quad (8)$$

$$\approx 0.008$$

with acoustic compliance $C_a = V/(\rho c^2) = 1.28 \times 10^{-9} \text{ m}^3/\text{Pa}$.

$\kappa \approx 0.008 \ll 1$: **weak coupling**. The gap area (4 mm^2) is tiny compared to the chamber area ($\sim 950 \text{ mm}^2$). But the Bernoulli mechanism **amplifies** the coupling. The chamber pressure fluctuation modulates the Bernoulli suction at the gap:

$$\beta = \frac{\delta p_{\text{chamber}}}{p_{\text{Bernoulli}}} = \frac{q_{\text{reed}} \cdot |Z_H(f_1)|}{\frac{1}{2} \rho v_{\text{gap}}^2} \approx 0.63 \% \quad (9)$$

This 0.63 % pressure modulation shifts the frequency by:

$$\Delta f_{\text{Bernoulli}} \approx f_1 \cdot \frac{\beta}{2} = 50 \times 0.003 = \mathbf{0.16 \text{ Hz}} \approx \mathbf{5.4 \text{ cents}} \quad (10)$$

Cross-references: Doc. 0002 · Doc. 0003

Pure acoustic coupling (κ^2) yields only 0.3 cents. But Bernoulli amplification ($\beta \approx 0.6\%$ pressure modulation) raises the frequency shift to **~5 cents** — clearly audible and consistent with practice. The two filters are not passively cascaded, but **actively coupled through the Bernoulli mechanism**.

4 The Complete Filter Combination

The closed transfer function of the total system:

$$H_{\text{ges}}(f) = \frac{H_1(f)}{1 - \kappa_{\text{eff}} \cdot H_1(f) \cdot H_2(f)}$$

(11)

The poles of H_{ges} give the actual oscillation frequencies of the coupled 4th-order system:

Mode 1 (reed mode):

$$f'_1 \approx f_1 \cdot \left(1 + \kappa^2 \frac{f_1^2}{f_H^2 - f_1^2}\right) = 50.000 \text{ Hz}$$

(12)

(+0.3 cents purely acoustic)

Mode 2 (chamber mode):

$$f'_2 \approx f_H \cdot \left(1 - \kappa^2 \frac{f_H^2}{f_H^2 - f_1^2}\right) = 273.98 \text{ Hz}$$

(13)

Mode repulsion: The modes repel each other — the lower is pushed down (after Bernoulli correction: *up*), the upper is pushed up. With Bernoulli amplification ($\beta = 0.63\%$), the shift of the reed mode becomes **~5 cents**.

5 Parametric Dependencies

Chamber Volume V

$\Delta k \sim 1/V$: smaller chamber $\rightarrow$ stiffer air spring $\rightarrow$ larger shift.

$V \text{ [cm}^3\text{]}$	$f_H \text{ [Hz]}$	$\Delta k \text{ [N/m]}$	Static [cents]	With Bernoulli [cents]
90	387	0.025	0.6	~11
120	335	0.019	0.4	~8
180	274	0.013	0.3	~5
240	237	0.009	0.2	~4
360	194	0.006	0.1	~3

Table 2: Frequency shift as a function of chamber volume

Gap Area A_{eff} (Deflection)

$\Delta k \sim A_{\text{eff}}^2$: more deflection $\rightarrow$ quadratically stronger coupling.

h_{max} [mm]	A_{eff} [mm ²]	Static [cents]	κ
0.5	1.3	0.03	0.003
1.0	2.7	0.13	0.005
1.5	4.0	0.29	0.008
2.0	5.3	0.52	0.010
2.5	6.7	0.81	0.013

Table 3: Frequency shift as a function of reed deflection (chamber 180 cm³)

Helmholtz Frequency f_H

Maximum coupling when $f_H = n \cdot f_{\text{reed}}$ (overtone exactly on resonance).

f_H [Hz]	Nearest OT	Distance [Hz]	G	Effect
150	3rd OT (150)	± 0	7.0	Strong - critical
200	4th OT (200)	± 0	7.0	Strong - critical
250	5th OT (250)	± 0	7.0	Strong - critical
274	5th OT (250)	+24	4.7	Noticeable
300	6th OT (300)	± 0	7.0	Strong - critical
325	6th OT (300)	+25	5.0	Noticeable
400	8th OT (400)	± 0	7.0	Strong - critical

Table 4: Amplification factor G at the nearest overtone. $f_H = 274$ Hz: $G = 4.7$ — not a direct hit, but close enough for noticeable coupling.

Optimisation rule: Choose f_H so that it lies **between two overtones**. The worst situation is $f_H = n \cdot f_{\text{reed}}$. For the 50 Hz reed with $f_H = 274$ Hz: 5th OT = 250 Hz and 6th OT = 300 Hz bracket f_H — no exact hit, favourable. This rule supplements the recommendation from Doc. 0002 Ch. 12 (avoiding resonant overtones) with the quantitative frequency-pulling aspect.

6 Block Diagram

Signal flow of the coupled filters:

[Bellows pressure] $\rightarrow$ [Valve] $\rightarrow$ $\underbrace{H_2(f) : \text{Chamber filter}}_{\text{Band-pass } f_H=274 \text{ Hz, } Q_H=7} \rightarrow$

[Gap coupler]
 $\rightarrow$ $\underbrace{H_1(f) : \text{Reed filter}}_{\text{Band-pass } f_1=50 \text{ Hz, } Q_1=50-150} \rightarrow$ [Sound radiation]

Feedback: $x \rightarrow S(x) \rightarrow q \rightarrow \delta p_{\text{chamber}} \xrightarrow{\beta=0.63\%} F_{\text{Bernoulli}} \rightarrow x$

Closed transfer function: $H_{\text{ges}}(f) = H_1(f)/(1 - \kappa_{\text{eff}} \cdot H_1 \cdot H_2)$

7 Practical Consequences

1. Tune in the chamber: The chamber load shifts frequency by 1-5 cents — more than the tuning tolerance ($\pm 1-2$ cents). Never tune on the bench.

2. Chamber changes require retuning: Any change to volume, partition wall (Doc. 0002 Ch. 8) or valve changes f_H , Q_H and thus the frequency shift.

3. Large chambers are more stable: $\Delta k \sim 1/V$ — large chambers are more tolerant of geometric variations.

4. Deflection acts quadratically: $\Delta k \sim A_{\text{eff}}^2$ — unevenly deflected reeds are harder to tune.

5. Push and draw reeds: Both see the same chamber impedance (Doc. 0002 Ch. 7, Doc. 0003 Ch. 3) $\rightarrow$ same shift $\rightarrow$ tremolo beat is preserved.

6. Temperature: $C_a = V/(\rho c^2)$, and $c^2 \sim T$. At $+10^\circ\text{C}$ the frequency shift decreases by $\sim 3\%$ — small compared to the direct temperature dependence of the reed.

8 Summary

The reed and chamber form a coupled two-filter system.

The reed is a narrow-band mechanical band-pass ($f_1 = 50$ Hz, $Q = 50$ -150); the chamber is a broad-band acoustic band-pass ($f_H = 274$ Hz, $Q = 5$ -10). Coupling occurs through the gap — nonlinearly through the Bernoulli mechanism (Doc. 0003), but linearisable for small-signal analysis.

Three mechanisms shift the frequency: (1) Static compliance load (+0.3 cents). (2) Resonant amplification at overtones near f_H (+1-5 cents). (3) Nonlinear feedback to the fundamental (+1-3 cents). **Total shift: typically 1-5 cents.**

The transfer function $H_{\text{ges}}(f) = H_1 / (1 - \kappa_{\text{eff}} \cdot H_1 \cdot H_2)$ is calculable. The dependencies (V , A_{eff} , f_H , Q_H) are quantitatively predictable. Agreement with practical findings confirms the model.

The reed determines the frequency — but the chamber has a say.

Dok. 0005

Frequency Shift

as Response Indicator



Frequency Shift as Response Indicator

Chamber Coupling, Operating Point and Geometric Influence

Proves that frequency shift and response deterioration are physically inseparable. Both are projections of the same complex coupling impedance $Z_H(f)$. Numerical values: 50 Hz bass reed from Doc. 0002.

Document References	
Doc. 0001	Analysis of instrument acoustics and psychoacoustics
Doc. 0002	Flow analysis of bass reed 50 Hz - v8
Doc. 0003	Impedance comparison: free-beating reed vs. labial pipe
Doc. 0004	Frequency variation - two-filter model
Doc. 0005	Frequency shift as response indicator (this document)
Doc. 0006	Symbol reference
Doc. 0007	Treble reed block - chamber frequencies
Doc. 0008	Tone modification by chamber geometry
Doc. 0500	Guide to aesthetic forensics of accordion casework
berechnungen_verifikation.py	Verification script for all calculations

1 The Central Thesis

Doc. 0002 (Ch. 10) shows: a resonant chamber deteriorates response through series coupling. Doc. 0004 shows: the chamber shifts the reed frequency by 1-5 cents. Both effects are caused by the same physical quantity — the **complex coupling impedance** $Z_H(f)$.

Frequency shift and response deterioration are the imaginary and real parts of the same complex coupling function. They are causally linked through the Kramers-Kronig relation. The frequency shift reveals whether response is improving or worsening — and how far one is from the critical resonance.

2 The Complex Coupling Impedance

Decomposition into Real and Imaginary Parts

The Helmholtz impedance of the chamber at frequency f :

$$Z_H(f) = R_H(f) + j X_H(f) \quad (1)$$

Real part (resistance = energy dissipation → response deterioration):

$$R_H(f) = Z_0 \cdot \frac{f/f_H}{Q_H \cdot D(f)} \quad (2)$$

Imaginary part (reactance = energy storage → frequency shift):

$$X_H(f) = Z_0 \cdot \frac{1 - (f/f_H)^2}{D(f)} \quad (3)$$

where $D(f) = (1 - (f/f_H)^2)^2 + (f/(f_H \cdot Q_H))^2$ and $Z_0 = 1/(\omega_H C_a)$.

Physical Assignment

Real part $R_H(f) \rightarrow$ **energy loss** → **response deterioration**:

$$\Delta\zeta_n = \kappa_{\text{eff}}^2 \cdot \frac{R_H(nf_1)}{Z_0}, \quad \Delta\tau_n = \frac{\Delta\zeta_n}{\pi f_1} \quad (4)$$

Imaginary part $X_H(f) \rightarrow$ **frequency shift**:

$$\Delta f_n = -\kappa_{\text{eff}}^2 \cdot f_n \cdot \frac{X_H(f_n)}{2Z_0} \quad (5)$$

$\Delta\tau$ (response) $\propto R_H(f)$ and Δf (frequency) $\propto X_H(f)$. Both come from **the same** complex impedance $Z_H = R_H + jX_H$. They relate to each other like the sine and cosine of an angle — know one, know the other.

3 The Sign Change as Warning Signal

The key insight lies in the **sign change** of the frequency shift:

- $\Delta f > 0$ (tone becomes **higher**): Nearest OT lies **below** f_H . Chamber acts as spring (compliance). Response is good, deteriorates as Δf grows.
- $\Delta f \rightarrow 0$ **with sign change**: OT passes through f_H . **Worst case**. Maximum energy extraction at minimum frequency signal.
- $\Delta f < 0$ (tone becomes **lower**): Nearest OT lies **above** f_H . Chamber acts as mass (inertance). Response was poor, improving.
- $\Delta f \approx 0$ **without sign change**: No OT near f_H . **Best case**.

The paradox: Exactly at optimal resonance ($f_H = n \cdot f_{\text{reed}}$), damping is maximal ($R_H = Q_H \cdot Z_0$), but the frequency shift **of that overtone** is zero ($X_H = 0$). At the resonance point, all coupling energy is converted to dissipation (real part); nothing remains for reactance (imaginary part). The overtone is not shifted but **damped** — which is worse.

Nyquist Diagram of the Coupling

In the Nyquist diagram (R_H horizontal, X_H vertical), the impedance traces a **circle**. Each overtone lies at a point on this circle:

OT n	R_H/Z_0	X_H/Z_0	Position on the circle
1 (50 Hz)	0.001	+1.000	↑ Upper edge (pure compliance)
3 (150 Hz)	0.011	+0.917	↑ Upper half
5 (250 Hz)	0.090	+0.537	↗ Approaching the peak
f_H (274)	0.143	≈ 0	→ Peak (maximum damping)
6 (300 Hz)	0.072	−0.374	↘ Below the peak
8 (400 Hz)	0.017	−0.655	↓ Lower half

Table 1: Nyquist positions of the overtones. At the peak (f_H): max. damping, X_H changes sign.

The 5th overtone (250 Hz) lies in the upper semicircle ($X > 0$, $\Delta f > 0$); the 6th overtone (300 Hz) in the lower ($X < 0$, $\Delta f < 0$). Between them, at $f_H = 274$ Hz, lies the peak. The fact that no overtone falls exactly on the peak explains the relatively

good response of the 50 Hz reed in this chamber.

4 The Mathematical Proof: Kramers-Kronig

The linkage is **physically compelling**. Every causal, linear response function satisfies the Kramers-Kronig relations:

$$X_H(\omega) = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{R_H(\omega')}{\omega' - \omega} d\omega' \quad (6)$$

$$R_H(\omega) = +\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{X_H(\omega')}{\omega' - \omega} d\omega' \quad (7)$$

($\mathcal{P}$ = Cauchy principal value). **Knowing the frequency dependence of the frequency shift (all $X_H(f)$), one can calculate the response deterioration (all $R_H(f)$) — and vice versa.**

For the Helmholtz resonator:

$$Z_H(f) = \frac{Z_0}{1 - (f/f_H)^2 + j f / (f_H \cdot Q_H)} \quad (8)$$

The real part (Lorentzian, bell shape) and imaginary part (dispersion curve, S-shape with zero crossing at f_H) are linked by the Hilbert transform — exactly like absorption and refractive index in optics.

The Kramers-Kronig relation is a consequence of **causality**. The chamber cannot “choose” to shift frequency without dissipating energy (or vice versa). Both effects are two sides of the same coin. A system that pulls the frequency strongly **must** also dissipate energy — and vice versa (except exactly at resonance, where $X = 0$).

5 The Practical Measurement Procedure

Step 1: Measure reed frequency **free** (reed plate on test block, no chamber). Result: f_{free} .

Step 2: Measure reed frequency **in the chamber** (same bellows pressure). Result: f_{chamber} .

Step 3: $\Delta f = f_{\text{chamber}} - f_{\text{free}}$.

Step 4: Interpretation:

Measurement result	Physics	Response
$\Delta f \approx 0$, response good	No OT near f_H	Optimal
$\Delta f > 0$ (tone higher)	Nearest OT below f_H	Good, deteriorating
Δf maximum positive	OT approaching f_H	Warning
$\Delta f \rightarrow 0$ (sign change)	OT passing through f_H	Worst case
$\Delta f < 0$ (tone lower)	Nearest OT above f_H	Improving
$\Delta f \approx 0$, response good	OT far from f_H	Optimal

Table 2: Interpretation key: $\Delta f \rightarrow$ response prognosis

Elegant variant: Adjustable chamber (sliding base). Vary volume V and thus f_H continuously. When Δf changes sign, one has passed through resonance — there response is worst. The synchrony of sign change and response minimum is the **experimental proof** of the Kramers–Kronig linkage.

6 Total Balance: Sum Over All Overtones

The total effect on the fundamental is the sum over all overtones:

$$\Delta f_{\text{total}} = \sum_n \frac{a_n}{n} \cdot \Delta f_n, \qquad \Delta \tau_{\text{total}} = \sum_n a_n^2 \cdot \Delta \tau_n \qquad (9)$$

The sign of the net frequency shift indicates from which side the dominant overtone approaches f_H :

Rule of thumb:

$\Delta f_{\text{net}} > 0 \rightarrow$ Dominant OT lies **below** $f_H \rightarrow$ lowering f_H worsens response.

$\Delta f_{\text{net}} < 0 \rightarrow$ Dominant OT lies **above** $f_H \rightarrow$ raising f_H worsens response.

$\Delta f_{\text{net}} \approx 0 \rightarrow$ Either decoupled (good) or on resonance (bad) $\rightarrow$ response test decides.

7 The Problem of the Missing Reference Frequency

The measurement procedure (f_{free} vs. f_{chamber}) has a fundamental **practical limitation** that does not appear in calculations but is critical for the instrument builder:

There is no absolute reference frequency. The frequency f_{free} (reed on test block, no chamber) is **itself** a setup-dependent quantity. The test block has its own volume, acoustic impedance, coupling. f_{free} is not the “true” natural frequency of the reed — it is the frequency of the reed **in a different acoustic environment**.

The physical natural frequency in vacuum (f_{vac}) is not practically measurable — the reed needs air to vibrate (Bernoulli mechanism). Even a reed in open space has an acoustic environment: free sound radiation as a weak load.

The Three “Reference Frequencies” of the Accordion Builder

$f_{\text{test block}}$: Reed on a standardised wooden block, blown with defined pressure. Closest to f_{vac} , but not identical (the block has compliance). Estimated difference from f_{vac} : 0.1–0.5 cents.

$f_{\text{chamber, open}}$: Reed mounted in chamber, valve open. The open valve changes chamber impedance compared to playing (lower Z at valve end $\rightarrow f_H$ shifts).

$f_{\text{chamber, playing}}$: Reed in chamber, valve closed, blown by bellows. This is the “real” operating state — but here the frequency is already modified by full chamber coupling.

The Pre-tuning Setup as Local Reference

Since none of the three reference frequencies is “absolute”, the experienced instrument builder works with a **local reference**: he defines his personal pre-tuning setup (specific test block, specific blowing pressure, specific room temperature) as the **zero point**. All frequency measurements are interpreted relative to this setup. The setup is the “calibration” of the tuning process.

The Δf method still works, because it measures a **difference**: $\Delta f = f_{\text{chamber}} - f_{\text{setup}}$. Absolute frequency is irrelevant — only the difference and above all the **sign change** when varying the chamber geometry. The sign change is setup-independent, because it is a property of the chamber-reed coupling, not of the absolute frequency.

What Experience Provides That Calculation Cannot

1. The offset correction: How many cents does f_{setup} lie above or below f_{vac} ? Only someone who has tuned with the

same test block for years and heard the resulting instruments can know.

2. Temperature compensation: f_{reed} changes with temperature (Young's modulus: $\sim 0.5 \text{ cents}/^\circ\text{C}$). c also changes ($+0.17 \text{ }^\circ\text{C} \rightarrow$ changes f_H and Δf). The experienced tuner compensates both intuitively.

3. Pressure dependence: Bellows pressure changes h_{eff} (Doc. 0002 Ch. 9) and thus S_{gap} , κ and Δf . The pre-tuning setup defines a specific operating point on the pressure curve.

4. Material variation: 0.1 mm difference in deflection $\rightarrow$ 7 % in S_{eff} (Doc. 0002 Ch. 13), and thus different κ and Δf . Calculation gives the average — experience says how far the individual case can deviate.

The role of calculation in the tuning process:

The formulas from Doc. 0002–0005 do not replace experience. They accomplish something different: they explain **why** the rules of experience work. They predict **in which direction** a change acts. They identify the **sensitive parameters**. They warn **when** caution is needed (sign change $\rightarrow$ resonance).

What calculation **cannot** do: predict the exact cent value, compensate for material variation, know the instrument builder's setup. For that, one needs the practical test — and the experience that makes craft from a thousand tests.

8 Clarification: Response vs. Loudness - No Trade-off in Chamber Tuning

A natural misconception must be addressed: if optimal response means the chamber draws as **little** energy as possible from the reed — does good response then reduce maximum achievable loudness?

No. The answer follows directly from the physics of series coupling (Doc. 0002 Ch. 10). Two effects must be clearly separated because they are physically distinct:

Chamber Tuning: Win-Win

When an overtone of the reed falls exactly on f_H , R_H is maximal — the chamber draws energy from the reed through series coupling. This energy loss deteriorates **both simultaneously**:

Response: The extracted energy is missing during amplitude build-up. Onset time τ increases by $\Delta\tau \propto R_H$ (Ch. 3).

Loudness: In the steady state, a fraction of the vibrational

energy **per cycle** flows into chamber dissipation instead of being converted to sound radiation. The equilibrium amplitude falls: $A_{\max} \propto 1/\sqrt{R_H}$.

Both effects stem from the same source — $R_H(f)$, the real part of the chamber impedance. When R_H falls (no overtone on f_H), **both** improve: faster response **and** larger amplitude. In chamber tuning there is no trade-off. Optimal tuning is win-win.

Reed Quality Factor Q_1 : The Real Trade-off

The trade-off between response and loudness exists — but it lies **not** in chamber tuning, but in the reed itself:

High Q_1 (low damping): Reed stores much energy per cycle → **louder**. But amplitude build-up takes many cycles → **slow response** ($\tau = Q_1/(\pi f)$).

Low Q_1 (high damping): Few cycles to reach steady state → **fast response**. But equilibrium amplitude is lower → **quieter**.

This trade-off is determined by material, clamping and deflection (Doc. 0002 Ch. 9) — **not** by chamber geometry.

Summary of the distinction:

Chamber tuning (f_H relative to overtones, Doc. 0005): Affects **external damping** by the chamber. Optimisation = minimise R_H = win-win (better response *and* more loudness).

Reed quality factor Q_1 (material, clamping, Doc. 0002): Affects **internal damping** of the reed. Here lies the real trade-off: higher Q_1 = louder but slower; lower Q_1 = faster but quieter. The instrument builder optimises both **independently**: tune the chamber to disturb as little as possible (f_H between overtones). Choose the reed so that Q_1 suits the musical context (bass: lower for fast response; treble: higher for loudness and sustain).

9 Summary

Proven relationship: Δf and $\Delta\tau$ are the imaginary and real parts of the same complex coupling impedance $Z_H(f)$, linked by Kramers-Kronig (causality).

Optimal resonance gives worst response: At $f_H = n \cdot f_{\text{reed}}$, R_H is maximal, $X_H = 0$. Maximum energy extraction at minimum frequency signal.

Frequency shift is a measurable indicator: Large $|\Delta f| \rightarrow$ strong coupling, but not critical. $\Delta f \rightarrow 0$ with sign change $\rightarrow$ **worst response**. $\Delta f \rightarrow 0$ without sign change $\rightarrow$ **best response**.

Measurement procedure: f_{free} vs. f_{chamber} . The sign change when varying chamber geometry marks the resonance crossing.

Missing absolute reference: f_{free} is setup-dependent. The pre-tuning setup (test block, blowing pressure, temperature) defines the local zero point. The sign change of Δf is setup-independent — it is the physical observable.

No trade-off in chamber tuning: Optimal tuning (f_H between overtones) improves response *and* loudness simultaneously, because both suffer from the same source (R_H). The trade-off between response and loudness lies in the reed quality factor Q_1 (material, clamping) — not in the chamber.

Calculation says where to look. Setup defines the zero point. Experience says what you find.

What you cannot measure, you cannot optimise. The frequency shift is the measurement quantity — the setup is the calibration.

Dok. 0006

Symbol Reference

Symbol Reference

Formula Symbols and Units of the Accordion Construction Series

Complete symbol reference for all formula symbols in Doc. 0001–0008. Organised by physical domain.

Doc. 0001	Instrument acoustics and psychoacoustics
Doc. 0002	Flow analysis of bass reed 50 Hz – v8
Doc. 0003	Impedance comparison: free-beating reed vs. labial pipe
Doc. 0004	Frequency variation – two-filter model
Doc. 0005	Frequency shift as response indicator
Doc. 0006	Symbol reference (this document)
Doc. 0007	Treble reed block – chamber frequencies
Doc. 0008	Tone modification by chamber geometry
Doc. 0500	Aesthetic forensics of accordion casework
berechnungen_verifikation.py	Verification script for all calculations

1 Geometric Quantities

Symbol	Unit	Description	Doc.
L	mm	Free reed length (70 mm)	0002
W	mm	Reed width (8 mm)	0002
t	mm	Reed thickness (0.354 mm)	0002
$h, h_{\max}$	mm	Tip deflection at free end (1.5 mm)	0002, 0004
h_{eff}	mm	Effective deflection under bellows pressure	0002
S_{gap}	mm ²	Effective gap area $= Wh_{\max}/3$	0002, 0004
S_{eff}	mm ²	Time-dependent flow area	0002
A_{eff}	m ²	Coupling area = S_{gap} in SI	0004, 0005
S_{valve}	mm ²	Valve opening area (244 mm ²)	0002
W_{slot}	mm	Slot width (9 mm)	0002
W_k	mm	Channel width (23.8 mm)	0002
V	cm ³	Chamber volume (180 cm ³)	0002–0005
V_{pre}	cm ³	Pre-chamber volume (67.2 cm ³)	0002
gap_{fold}	mm	Fold gap (19 mm)	0002
L_{eff}	mm	Effective acoustic length	0002, 0003
Δl	mm	Acoustic shortening by fold (≈ 15 mm)	0002, 0003
x	mm	Reed tip deflection	0002, 0004
A (amplitude)	mm	Vibration amplitude	0002

2 Material Properties

Symbol	Unit	Description	Doc.
E	Pa	Young’s modulus (steel: ~ 200 GPa)	0002
I	m^4	Second moment of area = $Wt^3/12$	0002
ρ_s	kg/m^3	Reed material density (7800)	0002, 0004
ρ	kg/m^3	Air density (1.2)	0002–0005
c	m/s	Speed of sound (343)	0002–0005
μ	Pa s	Dynamic viscosity of air	0002
η	–	Material loss factor	0002
m_{eff}	g	Effective mass, 1st mode ($0.243 \times m_{total}$)	0004
k_{mech}	N/m	Mechanical spring stiffness = $\omega_1^2 m_{eff}$	0004, 0005

3 Flow Quantities and Loss Coefficients

Symbol	Unit	Description	Doc.
Δp	Pa	Bellows pressure	0002
$\Delta p_{\min}$	Pa	Threshold pressure (≈ 224 Pa)	0002
Δp_{crit}	Pa	Critical pressure, gap closure (≈ 221 Pa)	0002
Δp_B	Pa	Bernoulli pressure = $\frac{1}{2}\rho v_{\text{tan}}^2$	0002
v_{gap}	m/s	Gap velocity = $\sqrt{2\Delta p/\rho}$	0002
v_{tan}	m/s	Tangential component at the gap	0002
Q (volume flow)	ml/s	Volume flow (~ 115 ml/s at 1 kPa)	0002
Re	–	Reynolds number	0002
ζ	–	Loss coefficient (general)	0002
ζ_{valve}	–	Valve loss (blow-through: 0.50; deflection: 1.70)	0002
ζ_{entry}	–	Slot entry loss (0.5)	0002
w_{tip}	mm	Static deflection under bellows pressure	0002

4 Directional Preservation

Symbol	Unit	Description	Doc.
α	°	Entry angle of the air jet	0002
β (wall angle)	°	Half-angle of angled partition (B: 3.7°)	0002
η_{total}	–	Total directional preservation factor	0002
η_{fold}	–	Preservation through 180° fold (≈ 0.6)	0002
η_{wall}	–	Preservation through partition (A: 0.3; B: 0.5; C: 0.7)	0002
f_{tan}	–	Tangential fraction = $(1 + \eta_{\text{total}})/2$	0002

5 Frequencies and Vibration Quantities

Symbol	Unit	Description	Doc.
f_1, f_{reed}	Hz	Mechanical natural frequency (50 Hz)	0002-0005
f_H	Hz	Helmholtz frequency of chamber (274 Hz)	0002-0005
$f_n = n f_1$	Hz	n -th overtone of the reed	0002, 0004, 0005
f_{free}	Hz	Reed frequency on test block	0005
f_{chamber}	Hz	Reed frequency in the chamber	0005
f'_1, f'_2	Hz	Coupled natural frequencies	0004
ω	rad/s	Angular frequency = $2\pi f$	0002-0005
n	-	Overtone number	0002-0005
λ	m	Wavelength = c/f	0002, 0003
Δf	Hz, cents	Frequency shift through coupling	0004, 0005

Unit cents: 1 cent = 1/1200 octave. Calculation: $\Delta f [\text{cents}] = 1200 \cdot \log_2(f_2/f_1)$. Musical perception threshold: $\pm 1\text{--}2$ cents.

6 Damping and Quality Factor

Symbol	Unit	Description	Doc.
Q, Q_1	-	Mechanical quality factor of reed (50-150)	0002, 0004, 0005
Q_H	-	Acoustic quality factor, Helmholtz (5-10)	0002-0005
Q_{total}	-	Total quality factor (harmonic sum)	0002
ζ (damping)	-	Damping ratio = $1/(2Q)$	0002, 0005
$\Delta\zeta_n$	-	Additional damping of OT n by chamber	0005
τ	ms	Onset time = $Q/(\pi f)$	0002, 0005
$\Delta\tau$	ms	Onset time extension	0005

7 Acoustic Impedance

Symbol	Unit	Description	Doc.
$Z_H(f)$	Pa s/m	Complex impedance, Helmholtz resonator	0002-0005
$Z_{\text{tube}}(f)$	Pa s/m	Input impedance of tube (organ pipe)	0003
$Z_{\text{acoust}}(t)$	Pa s/m	Time-variant gap impedance	0003
$Z_{\text{mech}}(f)$	N s/m	Mechanical impedance of reed	0003, 0004
Z_0	Pa s/m	Reference impedance $= 1/(\omega_H C_a)$	0005
$R_H(f)$	Pa s/m	Real part $Z_H \rightarrow$ damping/response	0005
$X_H(f)$	Pa s/m	Imaginary part $Z_H \rightarrow$ frequency shift	0005
C_a	m ³ /Pa	Acoustic compliance $= V/(\rho c^2)$	0004, 0005

8 Coupling Quantities (Two-Filter Model)

Symbol	Unit	Description	Doc.
$H_1(f)$	-	Transfer function, reed (band-pass f_1)	0004, 0005
$H_2(f)$	-	Transfer function, chamber (band-pass f_H)	0004, 0005
$H_{ges}(f)$	-	Closed transfer function = $H_1/(1 - \kappa_{eff}H_1H_2)$	0004, 0005
κ	-	Acoustic coupling constant (≈ 0.008)	0004
κ_{eff}	-	Effective coupling (with Bernoulli)	0004, 0005
β (Bernoulli)	-	Bernoulli amplification factor ($\approx 0.6\%$)	0004, 0005
Δk	N/m	Additional spring stiffness = $A_{eff}^2\rho c^2/V$	0004
$G(f)$	-	Impedance amplification near f_H (peak = Q_H)	0004, 0005
$D(f)$	-	Resonance denominator	0005
φ	°	Phase angle (Barkhausen: $\sum \varphi = 0$)	0004

9 Treble Reed Block (Doc. 0007)

Symbol	Unit	Description	Doc.
d (depth)	mm	Chamber depth (variable: 8→3 mm)	0007
d_{open}	mm	Diameter of circular opening (10 or 6 mm)	0007
S_{neck}	mm ²	Neck area = $\pi(d_{\text{open}}/2)^2$	0007
l_{neck}	mm	Physical neck height (8 mm)	0007
l_{eff}	mm	Effective neck length = $l_{\text{neck}} + 0.85 \cdot d_{\text{open}}$	0007
$p_{\text{convex}}(x)$	mm	Convex curve: $a \cdot (b/a)^x$ - below the straight line	0007
$p_{\text{concave}}(x)$	mm	Concave curve: $a + b - a \cdot (b/a)^{1-x}$ - above the line	0007
R_{eff}	-	Amplitude-weighted coupling load = $\sum a_n^2 R_{H,n}$	0007
a_n	-	Overtone amplitude $\approx 1/n$ (free-beating reed)	0007

10 Tone Modification (Doc. 0008)

Symbol	Unit	Description	Doc.
$A_n(f_H)$	-	Amplitude transfer of n -th OT: $1/(1 + \kappa_{\text{eff}}R_{H,n})$	0008
Φ_n	°	Phase shift of n -th OT by chamber	0008
Q_{rad}	-	Radiation quality factor (loss through opening)	0008
Q_{visc}	-	Viscous quality factor (wall friction): $2V/(\delta A_{\text{wall}})$	0008
Q_{turb}	-	Turbulence quality factor (at $\text{Re} \ll 1000$: $\rightarrow \infty$)	0008
δ	μm	Viscous boundary layer thickness = $\sqrt{2\mu/(\rho\omega)}$	0008
R_{rad}	Pa s/m^2	Acoustic radiation resistance of opening	0008
A_{wall}	mm^2	Total chamber interior surface area	0008

Dok. 0007

Treble Reed Block

Chamber Frequencies D3-C6

Treble Reed Block — Chamber Frequencies D3-C6

Seven Variants with Practical Depth Constraint

Complete analysis for 35 chambers (D3-C6). Seven variants: reference + three curve shapes each (linear, convex, concave) for depth and opening. Accounts for the practical constraint: chamber depth must not fall below the maximum reed amplitude. Amplitude-weighted evaluation ($a_n \sim 1/n$).

Doc. 0002	Flow analysis of bass reed 50 Hz - v8
Doc. 0004	Frequency variation - two-filter model
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1 Geometry and Starting Point

The treble reed block contains 35 chambers for the notes D3 (146.8 Hz) to C6 (1046.5 Hz). Each chamber is a rectangular box with constant width $W = 15.5$ mm and a length that decreases linearly from 50 mm (D3) to 30 mm (C6). The outlet opening is circular with $\varnothing 10$ mm ($S_{\text{neck}} = 78.5 \text{ mm}^2$) and a physical neck height of 8 mm (effective neck length $l_{\text{eff}} = 16.5$ mm with end correction $0.85 \cdot d$).

The Helmholtz frequency of each chamber depends on volume and opening:

$$f_H = \frac{c}{2\pi} \sqrt{\frac{S_{\text{neck}}}{V \cdot l_{\text{eff}}}} \quad (1)$$

From Doc. 0005 we know: response problems occur when the 2nd or 3rd overtone of a reed comes close to the chamber

7 Variants: A (Ref.) / B,C,Ci (Depth) / D,E,Ei (Opening)

resonance f_H . This corresponds to $f_H/f < 3$. Above $f_H/f = 4$ the chamber is acoustically “invisible” — coupling through higher overtones ($n \geq 5$) is energetically irrelevant ($a_n^2 < 4\%$).

2 The Practical Constraint: Reed Impact

The calculation shows that smaller chamber volumes push f_H upward, thereby improving response. But chamber depth cannot be reduced arbitrarily: **the reed must have room to vibrate**.

A free-beating reed swings down through the slot of the reed plate into the chamber at full volume. If the reed tip touches the chamber floor, the vibration is abruptly stopped — the reed “bottoms out”. This produces an ugly clicking sound, damages the reed over time, and limits loudness.

Maximum Reed Amplitude in the Treble

The maximum tip deflection depends on reed length and playing dynamics:

Note range	Reed length	Max. amplitude	Min. depth
D3 (lowest)	~35 mm	~3.5 mm	4.0 mm
A3	~28 mm	~2.8 mm	3.3 mm
A4 (middle)	~18 mm	~1.8 mm	2.3 mm
A5	~10 mm	~1.0 mm	1.5 mm
C6 (highest)	~8 mm	~0.8 mm	1.3 mm

Table 1: Estimated reed amplitudes and minimum depths (amplitude $\approx 10\%$ of reed length + 0.5 mm safety margin)

Minimum chamber depth must decrease from about 4 mm (D3) to 1.3 mm (C6). The variants calculated here with a minimum depth of 3 mm (C6) are **well above** the reed-impact limit.

Practical rule: Chamber depth should be as small as possible (minimum volume $\rightarrow$ highest $f_H \rightarrow$ best decoupling), but **never** below the maximum reed amplitude plus safety margin. For the lowest treble notes (D3–F3), 3 mm could be tight — 4 mm would be safer.

3 The Three Curve Shapes

Each geometric parameter (depth or opening diameter) can vary from its starting value a to its final value b in three different

7 Variants: A (Ref.) / B,C,Ci (Depth) / D,E,Ei (Opening)

ways:

$$\text{Linear: } p(x) = a + x(b - a) \quad (2)$$

$$\text{Convex: } p(x) = a \cdot (b/a)^x \quad (3)$$

$$\text{Concave: } p(x) = a + b - a \cdot (b/a)^{1-x} \quad (4)$$

All three connect **the same endpoints**: $p(0) = a$, $p(1) = b$. They differ in the **distribution** of the reduction across the note range:

Linear distributes the reduction evenly: each semitone step changes the parameter by the same absolute amount (e.g. -0.147 mm/semitone for $8 \rightarrow 3$ mm).

Convex (lies **below** the straight line) reduces faster at the beginning and slower at the end. In the middle range the parameter has a **lower** value than the linear curve. Example depth: at A4, convex gives 4.9 mm vs. linear 5.5 mm — 0.6 mm less.

Concave (lies **above** the straight line) reduces slower at the beginning and faster at the end. In the middle range the parameter has a **higher** value than the linear curve. Example depth: at A4, concave gives ≈ 6.1 mm vs. linear 5.5 mm — 0.6 mm more.

Which curve is better depends on which direction the parameter shifts f_H :

If **less** of the parameter **raises** f_H (as with depth: less $V \rightarrow$ higher f_H), the curve that has less in the middle range is better $\rightarrow$ **convex wins**.

If **more** of the parameter **raises** f_H (as with opening: more $S \rightarrow$ higher f_H), the curve that has more in the middle range is better $\rightarrow$ **concave wins**.

4 The Seven Variants

Var.	Short name	Depth [mm]	Ø [mm]	Opening	Effect on f_H
A	Reference	8 constant	10 constant		Reference
B	d linear	8→3 linear	10 constant		f_H rises (V falls)
C	d convex	8→3 convex	10 constant		f_H rises more
Ci	d concave	8→3 concave	10 constant		f_H rises less
D	Ø linear	8 constant	10→6 linear		f_H falls (S falls)
E	Ø convex	8 constant	10→6 convex		f_H falls more
Ei	Ø concave	8 constant	10→6 concave		f_H falls less

Table 2: The seven variants. Upper group: depth varies. Lower group: opening varies.

Why Direction Decides

Reducing depth decreases chamber volume V . Since $f_H \propto 1/\sqrt{V}$, f_H **rises**. This shifts f_H/f upward — away from the low overtones. This is the **right direction**.

Reducing opening decreases neck area S_{neck} and also the end correction ($l_{\text{eff}} = 8 + 0.85d$ also falls). Net effect: $f_H \propto \sqrt{S/l_{\text{eff}}}$ **falls**. Quantitatively: Ø from 10 to 6 mm lowers f_H to **67%**. This shifts f_H/f **downward** — the **wrong direction**.

5 Main Result

	A	B	C	Ci	D	E	Ei
f_H at D3 [Hz]	1513	1513	1513	1513	1513	1513	1513
f_H at C6 [Hz]	1953	3189	3189	3189	1315	1315	1315
f_H/f min.	1.9	3.0	3.0	3.0	1.3	1.3	1.3
Critical ($OT \leq 2$)	7	0	0	0	12	13	12
Noticeable ($OT = 3$)	7	7	5	8	5	5	5
Good ($OT \geq 4$)	21	28	30	27	18	17	18
$\sum R_{\text{eff}}$	17.4	12.8	12.3	13.5	17.2	17.3	17.0

Table 3: Main comparison. Bold: variant C (best result).

Ranking

1. **C** (depth convex): 0 crit., 5 noticeable, 30 good. $\sum R = 12.3$
2. **B** (depth linear): 0 crit., 7 noticeable, 28 good. $\sum R = 12.8$

7 Variants: A (Ref.) / B,C,Ci (Depth) / D,E,Ei (Opening)

3. **Ci** (depth concave): 0 crit., 8 noticeable, 27 good. $\sum R = 13.5$
4. **Ei** ($\emptyset$ concave): 12 crit., 5 noticeable, 18 good. $\sum R = 17.0$
5. **D** ($\emptyset$ linear): 12 crit., 5 noticeable, 18 good. $\sum R = 17.2$
6. **E** ($\emptyset$ convex): 13 crit., 5 noticeable, 17 good. $\sum R = 17.3$
7. **A** (reference): 7 crit., 7 noticeable, 21 good. $\sum R = 17.4$

6 Analysis of the Depth Group

All three depth variants (B, C, Ci) eliminate critical resonances completely (0 critical notes). They differ only in the number of noticeable resonances:

C (convex, below the straight line): 5 noticeable. The curve gives the middle notes less volume $\rightarrow$ higher $f_H \rightarrow$ better decoupling. Example A4: depth 4.9 mm, $f_H/f = 5.1$. **Best depth variant.**

B (linear): 7 noticeable. Uniform reduction. A4: depth 5.5 mm, $f_H/f = 4.8$.

Ci (concave, above the straight line): 8 noticeable. The curve retains more volume in the middle range $\rightarrow$ lower f_H . Worst depth variant — but still 0 critical.

The difference between the three depth variants is **moderate** (12.3 vs. 12.8 vs. 13.5 in $\sum R_{\text{eff}}$). The **big jump** is from reference A (17.4) to any of the three depth variants — depth reduction itself is much more important than the curve shape.

Practical significance: The difference between convex and linear is 2 fewer noticeable resonances — audible, but not dramatic. The difference between depth reduction and none (A vs. B/C/Ci) is 7 critical resonances — this is the decisive step. **Reducing the depth achieves 95% of the benefit. Curve shape is fine-tuning.**

7 Analysis of the Opening Group

The three opening variants (D, E, Ei) make the situation worse compared to the reference — they have **more** critical notes (12–13 vs. 7). The reason: $\emptyset$ from 10 to 6 mm reduces S_{neck} from 78.5 to 28.3 mm² (–64 %) and l_{eff} from 16.5 to 13.1 mm (–21 %). Net effect: f_H falls to **67 %**. f_H/f drops to 1.3 at C6 — there f_H lies **below** the reed frequency itself.

Ei (concave): 12 crit. — least harmful, because the curve retains the larger openings in the middle range.

E (convex): 13 crit. — most harmful, because the curve reduces the opening most in the middle range.

Conclusion, opening group: None of the three opening variants is better than reference A. Reducing the opening from 10 to 6 mm alone is **counterproductive**. Even the best opening variant (Ei) has 12 critical notes — nearly twice as many as the reference (7). Varying the opening alone is the wrong approach.

8 Minimise Volume — The Practical Optimisation Rule

The analysis yields a clear optimisation rule, consistent with the practical experience of instrument builders:

Optimisation rule: volume as small as the reed allows.

For each chamber, depth is chosen such that:

1. The reed does **not bottom out** at maximum volume (depth $\geq$ max. amplitude + 0.5 mm safety margin).
2. Volume is **no larger** than required by condition (1).
3. The opening remains **as large as possible** (10 mm), because any reduction pushes f_H in the wrong direction.

In practice: chamber depth follows reed amplitude — short reeds (high notes) need less space, receive shallower chambers and automatically benefit from higher f_H/f . The convex depth profile (C) is slightly better than linear (B) because it reduces more aggressively in the middle range.

The result agrees with instrument-building practice: experienced builders make chambers just deep enough for the reed not to bottom out, and no deeper. The calculation now shows **why** this works: less volume $\rightarrow$ higher $f_H \rightarrow$ fewer overtone resonances $\rightarrow$ better response.

9 Summary

Seven variants for the treble reed block D3-C6:

Reducing depth (B, C, Ci) raises f_H — right direction, 0 critical notes.

Reducing opening (D, E, Ei) lowers f_H — wrong direction, 12-13 critical.

Within depth: **convex** > **linear** > **concave** (12.3 vs. 12.8 vs. 13.5).

Within opening: **concave** > **linear** > **convex** (17.0 vs. 17.2 vs. 17.3).

In both cases, the curve wins that retains more of the parameter that pushes f_H upward.

The big jump lies in depth reduction itself (7 fewer critical), not in the curve shape (2 fewer noticeable). Reducing depth achieves 95 % of the benefit.

Practical rule: Chamber depth = maximum reed amplitude + 0.5 mm. No deeper. Opening as large as possible.

Keep $f_H/f \geq 3 \rightarrow$ no 2nd OT hit $\rightarrow$ no critical problems.

Minimum volume, maximum opening — the chamber as small as the reed allows.

Dok. 0008

Tone Modification

by Chamber Geometry



Tone Modification by Chamber Geometry

The Chamber as Notch Filter in the Overtone Spectrum

Analyses how chamber geometry (volume, opening) changes the timbre. The chamber acts as a frequency-dependent notch filter: it attenuates overtones near f_H and shifts their phase. Both effects modify the waveform and thus the perceived sound.

Doc. 0002	Flow analysis of bass reed 50 Hz - v8
Doc. 0005	Frequency shift as indicator of response
Doc. 0007	Treble reed block - chamber frequencies (7 variants)
Doc. 0008	Tone modification by chamber geometry (this document)
berechnungen_verifikation.py	Verification script for all calculations

1 The Chamber as Notch Filter — Not an Amplifier

A common misconception: the chamber “amplifies” the tone, like the resonating body of a guitar. **The opposite is true.** Due to series coupling (Doc.0002 Ch.10, Doc.0003), the chamber draws energy from the reed at resonance — it damps rather than amplifies.

The chamber acts on the overtone spectrum like a **notch filter**: it creates a “hole” in the spectrum at f_H . The transfer function for the n -th overtone:

$$A_n(f_H) = \frac{1}{1 + \kappa_{\text{eff}} \cdot R_H(f_n)} \quad (1)$$

At $f_n = f_H$, R_H is maximal and attenuation is strongest. Far from f_H , $R_H \approx 0$ and $A_n \approx 1$ (unattenuated).

The chamber is not a resonating body, but a notch filter. It creates a hole in the overtone spectrum at f_H . The reed itself generates all overtones — the chamber filters some of them out. Which overtones are affected depends on f_H (volume and opening, Doc.0007). How strongly they are affected depends on the coupling strength κ_{eff} .

2 Amplitude Attenuation and Phase Shift

Doc. 0005 shows: the real part $R_H(f)$ and imaginary part $X_H(f)$ of the chamber impedance are linked by the Kramers–Kronig relations. For the sound: **every amplitude attenuation is accompanied by a phase shift.**

Amplitude Attenuation (Real Part R_H)

$R_H(f)$ has the form of a Lorentzian curve, centred at f_H with half-width f_H/Q_H . Overtones within this width are reduced in amplitude:

Affected OT	Frequency-range	Tonal effect
$n = 2$ (octave)	$2f$	Tone becomes thin, hollow - 25 % energy missing
$n = 3$ (twelfth)	$3f$	Tone becomes nasal, throaty - 11 % missing
$n = 4$ (double octave)	$4f$	Tone loses clarity - 6 % missing
$n = 5\text{--}6$	$5\text{--}6f$	Tone loses brilliance - barely noticeable
$n \geq 7$	$\geq 7f$	Barely audible - amplitude < 2 % anyway

Table 1: Tonal effect of overtone attenuation by overtone order

Phase Shift (Imaginary Part X_H)

$X_H(f)$ has the form of an S-curve (dispersion curve): below f_H the phase is shifted **lagging**, above it **leading**:

$$\Phi_n = \arctan\left(Q_H \cdot \left(\frac{f_n}{f_H} - \frac{f_H}{f_n}\right)\right) \tag{2}$$

The phase shift modifies the **waveform** of the tone without significantly changing individual amplitudes. The ear perceives this as a change in **sharpness** or **softness**. The effect is strongest when neighbouring overtones experience different phase shifts — this distorts the waveform most noticeably.

The phase shift explains why experienced instrument makers hear tonal differences that a spectrum analyser barely shows. Amplitudes change by only a few percent, but phase relationships change significantly — and this reshapes the waveform.

3 What the Variation Factors Change in the Sound

Reducing Depth (Variants B, C, Ci): f_H rises → notch shifts upward

When depth drops from 8 to 3 mm, f_H rises from 1500–1950 Hz to 1500–3190 Hz. The notch filter hole shifts from the 3rd–5th overtones up to the 5th–8th overtones.

Tonal effect: The energy-rich low overtones (2nd, 3rd) are **released**. The tone becomes **fuller, richer in fundamental, warmer**. Brilliance is slightly reduced — but the high overtones carry little energy anyway ($a_5^2 = 4\%$). The loss is barely audible.

Reducing Opening (Variants D, E, Ei): f_H falls → notch shifts downward

When $\varnothing$ drops from 10 to 6 mm, f_H falls to 67 %. The notch hole shifts into the range of the 2nd–3rd overtones.

Tonal effect: The energy-rich low overtones are **attenuated**. The tone becomes **thin, hollow, lifeless**. The 2nd overtone carries 25 % of the total energy — when it is attenuated, the tone loses its “body”. This is not just a deterioration of response, but also a **tonal impoverishment**.

Q_H : The Width of the Notch Filter

Q_H determines how narrow the notch filter hole is. Quantitative analysis of the loss mechanisms shows that Q_H is determined almost entirely by the **radiation losses at the opening** — not by the wall shape:

	Q_{rad}	Q_{visc}	Q_{tot}	Dominant loss
Treble (5 cm ³ , 1691 Hz)	43	88	29	Radiation (67 %)
Bass (180 cm ³ , 274 Hz)	263	113	79	Wall friction (70 %)

Table 2: Decomposition of Q_H . Wall shape changes Q_{visc} by $\pm 5\text{--}10\%$ (area), shifting Q_{total} by at most $\pm 3\%$ (treble) to $\pm 7\%$ (bass) — inaudible.

High Q_H (large opening, little radiation): **Narrow, deep hole.** A single overtone is strongly attenuated. Musically: **sharp tonal colouring.**

Low Q_H (small opening, much radiation): **Wide, shallow hole.** Several overtones lightly attenuated. Musically: **diffuse dullness.**

Q_H is determined almost entirely by the **opening geometry** (diameter, neck length), not by the wall shape. The audible difference between different partition wall shapes (Doc.0002 practical finding) does not run through Q_H ($< 7\%$ change), but through the **unsteady impulse response** — reflection patterns, position of the acoustic end, pressure pulse transport. These effects are real, but not calculable in the Helmholtz model.

4 Tonal Character Across the Note Range

Lower third (D3-C#4): f_H far above the low overtones. Chamber “invisible”. Tone full and natural. No difference between variants.

Middle third (D4-C#5): For variant A, the 3rd-5th OT approach f_H . Slight attenuation — tone becomes somewhat “nasal”. For C the attenuation is less.

Upper third (D5-C6): For A, the 2nd OT hits f_H — the hole sits at the octave. Tone becomes **noticeably thinner**. For C, f_H lies at the 3rd-4th OT — tone remains fuller.

5 The Three Tonal Dimensions

Dimension	Physics	Determined by	Perceived sound
Position of the hole	f_H	Volume V , opening S	Which OT is missing
Width of the hole	Q_H	Opening geometry (67-70 %)	Sharp vs. diffuse
Depth of the hole	κ_{eff}	Gap area, volume	Strong vs. weak

Table 3: The three tonal dimensions. Position is the dominant lever.

6 Response, Loudness and Tone: The Same Optimum

Doc.0005 (Ch. 10) shows: optimal chamber tuning improves response *and* loudness (win-win through R_H). This document adds: **the tone also benefits.**

All three suffer from the same source: $R_H(f)$. It draws energy ($\rightarrow$ response, loudness) **and** attenuates overtones ($\rightarrow$ tone). When R_H is minimised, all three improve.

There is no trade-off between response, loudness and tone in chamber optimisation. All three are served by the same optimum: $f_H/f \geq 3$, minimal volume, large opening. The only real trade-off lies in the **reed quality factor** Q_1 : louder and richer in overtones, but slower. That is a reed decision, not a chamber decision.

7 Summary

The chamber is not a resonating body, but a notch filter. It creates a hole in the overtone spectrum at f_H .

Which overtone is affected determines the tonal character:

- 2nd OT attenuated ($f_H/f \approx 2$): **thin, hollow** - 25 % energy missing
- 3rd OT attenuated ($f_H/f \approx 3$): **nasal** - 11 % missing
- 5th+ OT attenuated ($f_H/f \geq 5$): **barely changed** - < 4 % energy

Three tonal dimensions: Position (f_H via volume/opening — dominant lever), width (Q_H via opening geometry — wall shape < 7 %), depth (κ_{eff} via gap area).

The phase shift (Kramers-Kronig) modifies the waveform — even when amplitudes appear barely changed in a spectrum analyser. This explains why trained ears hear differences that instruments do not show.

Response, loudness and tone benefit from the same optimum: $f_H/f \geq 3$, minimal volume, large opening. No trade-off in chamber optimisation.

The chamber does not create the sound — it filters it. Less filter, more music.

Dok. 0009

Frequency Coupling

of Multiple Reeds

Frequency Coupling of Multiple Reeds

Locking, Pulling and the Musette Effect

When two or three reeds tuned with tremolo detuning ($\approx 1\text{ Hz}$) sound into the same chamber or through the same opening, they influence each other's frequency. The extent depends on how strongly the chamber resonance couples the overtones. This document presents an explanatory model — the three-zone gradient model — that traces the observed phenomena (locking, ghost peaks, response loss) back to the chamber impedance in a physically plausible way. Numerical values are order-of-magnitude estimates based on simplified models and empirical calibration — they provide qualitative understanding, not constructive specifications. Values refer to the treble reed block from Doc. 0007, variant C (convex).

Doc.	Title
0002	Flow analysis of bass reed 50 Hz — v8
0004	Frequency variation — two-filter model
0005	Frequency shift as response indicator
0007	Treble reed block — chamber frequencies (7 variants)
0008	Tone modification by chamber geometry

1 The Practical Observation

When tuning tremolo registers, the accordion builder observes three different effects that depend on the note range:

Range	f_H/f	Observation
Low notes (D3-G#3)	> 5	Two or three reeds tuned to ≈ 1 Hz beat lock onto a single frequency when sharing an opening. The tremolo disappears. With separate air supply, the effect does not occur.
Middle notes (A3-C#5)	$3\text{--}5$	The tremolo is audible, but the spectrum analyser shows additional ghost peaks beside the three expected peaks at equal spacing. Instead of three peaks, four or five appear.
High notes (D5-C6)	$2\text{--}3$	The tremolo works, but response is sluggish and the tone thin. Coupling is strong, but manifests as energy loss, not frequency shift.

The key observation: these three effects are **not** binary (present/absent), but form a **gradient field**. Transitions are gradual, and a note can show characteristics of two zones simultaneously. The effects do not occur at exactly the notes whose overtones hit the chamber resonance, but at notes that come **close** to the critical coupling.

2 The Coupling Mechanism — Chamber Resonance as Mediator

Coupling between multiple reeds runs **not** through the weak compliance air spring (shared chamber volume as spring), but through the **chamber resonance** itself. The mechanism is the same that explains response deterioration in Doc. 0005:

Step 1 — Overtone hits chamber resonance: Reed A vibrates at f_A and produces overtones $n \cdot f_A$. When an overtone lies close to f_H , it is amplified by the chamber resonance (impedance amplification G , Doc. 0004 and 0007).

Step 2 — Resonantly amplified pressure couples to Reed B: The amplified overtone creates a pressure excess in the chamber that Reed B “sees” as a periodic force. This force is proportional to $G(n \cdot f_A)$ and the overtone amplitude $a_n \propto 1/n$.

Step 3 — Nonlinear amplification at full amplitude: At full volume, the effective gap area A_{eff} is not 2 mm^2 (rest

position) but time-averaged $\approx 6 \text{ mm}^2$ (Doc. 0002 Ch. 8: gap opens $8\times$ per cycle). Coupling strength scales with A_{eff}^2 , giving a factor $(6/2)^2 = 9$ over the linear model.

Why the Linear Model Fails

The Helmholtz model from Doc. 0002–0008 calculates compliance coupling and obtains $\Delta f_{\text{lock}} < 0.01 \text{ Hz}$ — far too weak. Practice shows 0.5–1 Hz. The missing factors are: (a) resonant amplification $G \approx 3\text{--}7$ at overtones near f_H , and (b) the nonlinear gap amplitude at full volume (factor 9). Together: $0.01 \times 5 \times 9 \approx 0.5 \text{ Hz}$ — consistent with observation.

Core statement: Coupling between reeds is mediated by the same chamber impedance $Z_H(f)$ that determines response in Doc. 0005 and tone in Doc. 0008. Response, tone and tremolo coupling are three projections of the same physical quantity.

3 The Adler Equation — Injection Locking

The physics of coupled oscillators with small frequency differences is described by the Adler equation (1946). It applies universally to lasers, electronic oscillators, biological rhythms — and reeds:

$$\frac{d\varphi}{dt} = 2\pi \Delta f - 2\pi \Delta f_{\text{lock}} \cdot \sin(\varphi) \quad (1)$$

Here Δf is the tuned frequency difference and Δf_{lock} is the lock-in range.

Regime 1 — Locking ($|\Delta f| < \Delta f_{\text{lock}}$)

The phase φ converges to a fixed value. Both reeds vibrate at the same frequency. The tremolo disappears completely.

Regime 2 — Pulling ($|\Delta f| > \Delta f_{\text{lock}}$)

The reeds vibrate at different frequencies, but the actual beat is less than tuned:

$$\Delta f_{\text{eff}} = \sqrt{\Delta f^2 - \Delta f_{\text{lock}}^2} \quad (2)$$

For a 1 Hz tremolo with $\Delta f_{\text{lock}} = 0.7 \text{ Hz}$ (shared opening): $\Delta f_{\text{eff}} = \sqrt{1^2 - 0.7^2} = 0.71 \text{ Hz}$ — tremolo slows by 30 %. With separate openings ($\Delta f_{\text{lock}} \approx 0.1 \text{ Hz}$): $\Delta f_{\text{eff}} = 0.995 \text{ Hz}$ — practically unchanged.

Three Reeds (Musette)

With three reeds at $\Delta f = \pm 1$ Hz, the effect intensifies. The middle reed “sees” two coupling sources and the effective lock-in range grows by factor $\approx \sqrt{2}$: $\Delta f_{\text{lock},3} \approx 1.4 \cdot \Delta f_{\text{lock},2}$. With a shared opening this can be enough to lock all three reeds.

4 The Three-Zone Gradient Model

The coupling strength Δf_{lock} is not the same for all notes. It depends on how the overtones of each note relate to the chamber resonance f_H . Three zones with different dominant effects result:

Zone 1: Locking Zone ($f_H/f > 5$, low notes)

For the lowest notes (D3–G#3, $f = 147$ –208 Hz), f_H falls at the 7th–12th overtone. Each individual overtone couples weakly (energy fraction $a_n^2 = 1$ –4%), but the overtone grid is **dense**: the spacing between adjacent overtones is only f_0 (147–208 Hz), and the resonance bandwidth $f_H/Q_H \approx 200$ –300 Hz is wide enough that **always** at least one overtone lies in the band.

The sum of many weak couplings yields moderate total coupling. At full volume (nonlinear factor $\times 9$) this is sufficient to drive Δf_{lock} to 0.5–1.5 Hz — enough for locking at 1 Hz tremolo. Subjectively these are the notes where the tremolo “locks in” or “blurs”.

Zone 2: Ghost Peak Zone ($f_H/f \approx 3$ –5, middle notes)

In the middle of the range (A3–C#5, $f = 220$ –554 Hz), the 3rd–5th overtone hits f_H . Each of these overtones carries 4–11% energy — significantly more than in Zone 1. But the overtone grid is wider (spacing 220–554 Hz), so only one or two overtones lie in the resonance band simultaneously.

Coupling is strong enough for **intermodulation**, but not for locking. The spectrum shows mixing products: when Reed A vibrates at f_A and Reed B at $f_B = f_A + 1$ Hz, the nonlinear coupling through the chamber resonance generates sidebands at $f_A \pm 2$ Hz and $f_B \pm 2$ Hz. With three reeds ($f - 1$, f , $f + 1$), peaks appear at $f - 2$ and $f + 2$ — the observed “fourth and fifth peaks”.

Zone 3: Response Zone ($f_H/f \approx 2$ –3, high notes)

In the upper third (D5–C6, $f = 587$ –1047 Hz), the 2nd or 3rd overtone falls on f_H . These overtones carry 11–25% of total energy

— massive coupling. But the effect manifests primarily as **energy loss** (R_H maximum, Doc. 0005), not frequency shift. The reed has a slow onset and sounds thin.

Tremolo pulling is strongest, but secondary: the note already responds poorly, and at reduced amplitude the nonlinear amplification factor falls. In practice, the tuner notices the poor response rather than tremolo instability.

Table 1: Overview of the three zones

Zone	Notes	f_H/f	Dom. OT	Energy	Mechanism	Remedy
1 Locking	D3-G#3	> 5	7-12	1-4 %	Many weak OT sum. Locking at full volume.	Separate openings
2 Ghost peaks	A3-C#5	3-5	3-5	4-11 %	Individual medium-strength OT generate intermodulation.	Tune tremolo wider
3 Response	D5-C6	2-3	2-3	11-25 %	Few strong OT → energy loss dominates over Δf .	Reduce chamber depth

Gradual boundaries: The boundaries between zones are not sharp. A note like A3 (Zone 1/2 boundary) may still show locking with a shared opening, but ghost peaks with separate openings. The zone depends not only on f_H/f , but also on coupling strength (shared vs. separate) and playing volume (nonlinear factor).

5 Diagrams

(See PDF version, generated with `build_0009_frequenzkopplung.py`: 5 diagrams — zone overview, ratio/overtone, Adler pulling, coupling landscape, spectrum comparison.)

6 Shared vs. Separate Openings — Quantitative Comparison

Practice clearly shows: separate openings (each reed gets its own air supply with some spacing) drastically reduce all three coupling effects.

With a shared opening, the reeds share the same acoustic resistance at the neck. The amplified overtone from Reed A creates pressure in the chamber that Reed B “sees” directly — the chamber is the coupling medium. With separate openings, pressure from Chamber A must travel through the bellows (large volume, very low impedance) to Chamber B. The bellows volume (several litres) acts as a low-pass filter with cut-off fre-

quency far below 1 Hz — acoustic coupling is weakened by factor ≈ 7 .

Table 2: Empirical comparison values: shared vs. separate openings (full playing volume)

Parameter	Shared	Separate	Factor
Δf_{lock} (2 reeds)	≈ 0.7 Hz	≈ 0.1 Hz	7×
1 Hz tremolo becomes	0.71 Hz (−29 %)	0.995 Hz (−0.5 %)	—
Δf_{lock} (3 reeds, musette)	≈ 1.0 Hz	≈ 0.14 Hz	7×
Ghost peaks at A4	−24 dB	−41 dB	17 dB
Locking boundary note	$\approx$ G#3 (208 Hz)	$\approx$ D3 (147 Hz)	—

7 Practical Consequence — Tuning Correction

When separate openings are not possible (design, space), the tremolo must be tuned **deliberately wider** to compensate for the pulling effect:

$$\Delta f_{\text{tuning}} = \sqrt{\Delta f_{\text{target}}^2 + \Delta f_{\text{lock}}^2} \tag{3}$$

For a desired tremolo of 1 Hz with $\Delta f_{\text{lock}} = 0.7$ Hz:

$$\Delta f_{\text{tuning}} = \sqrt{1^2 + 0.7^2} = \sqrt{1.49} = 1.22 \text{ Hz} \tag{4}$$

One tunes 22 % wider than desired. This correction value depends on the note — larger in Zone 1 (low notes), smaller in Zone 2 (middle notes). In Zone 1 the correction can become so large that locking is unavoidable — then only separate openings or a change in chamber geometry remain.

Table 3: Tuning compensation table (shared opening, full volume)

Note	Δf_{target} [Hz]	Δf_{lock} [Hz]	Δf_{tuning} [Hz]	Note
D3 (147 Hz)	1.0	~ 1.2	1.56	Locking possible
A3 (220 Hz)	1.0	~ 0.8	1.28	Strong pulling
A4 (440 Hz)	1.0	~ 0.5	1.12	Moderate pulling
D5 (587 Hz)	1.0	~ 0.6	1.17	Pulling + response
A5 (880 Hz)	1.0	~ 0.4	1.08	Weak pulling

8 Connection to Single-Reed Physics (Doc. 0005)

The correlation between tremolo coupling and response is not coincidental — it follows from the physics. Both effects are
Three-Zone Gradient Model

governed by the same complex chamber impedance $Z_H(f) = R_H + jX_H$ (Doc. 0005):

Real part R_H : Determines energy loss. For the single reed this means poorer response (Doc. 0005). For multi-reed coupling it means: the chamber absorbs and re-emits energy — exactly the mechanism that mediates coupling. **Large R_H = poor response AND strong coupling.**

Imaginary part X_H : Determines frequency shift. For the single reed, X_H shifts the overtones (Doc. 0005). For multi-reed coupling, X_H determines the phase relationship between coupled reeds — and thus whether pulling or locking occurs.

Since R_H and X_H are linked by the Kramers-Kronig relation (Doc. 0005 Ch. 3): **a note that has response problems (large R_H at an overtone) simultaneously has strong tremolo pulling.** But the effect manifests differently: response deteriorates exactly at $f_H = n \cdot f$ (R_H maximum, $X_H = 0$). Pulling is maximum **just beside** the resonance (X_H maximum, R_H falling). Therefore the strongest tremolo effects do not occur at exactly the same notes as the worst response — but at the neighbouring notes. This is precisely the observation.

9 Why the Zones Show Different Effects

The different manifestations in the three zones follow from the structure of the overtone grid:

Zone 1 (low notes) → Locking: The overtone grid is dense (spacing = $f_0 \approx 150$ Hz). At $f_H \approx 1500$ Hz, ~ 10 overtones are within reach, at least one always in the resonance band. Coupling is broadband — it acts at any detuning, not only at specific Δf values. This favours locking, because the coupling force has no “gaps”.

Zone 2 (middle notes) → Ghost peaks: The overtone grid is wider (spacing ≈ 300 – 500 Hz). Coupling is narrowband — it acts through only one or two overtones that fall in the resonance band. This selective coupling produces intermodulation products at discrete frequencies (sidebands), rather than broadband synchronisation. The result is ghost peaks in the spectrum, not locking.

Zone 3 (high notes) → Response: The 2nd overtone (octave) lies on or near f_H . This overtone carries 25 % of total energy — coupling is maximal, but manifests as massive energy

absorption by the chamber (R_H dominates). The reed loses so much energy to the chamber resonance that amplitude falls. At reduced amplitude the nonlinear amplification factor also falls — the vicious circle favours energy loss over frequency shift.

10 Limits of the Model

The three-zone model explains the qualitative observations and provides quantitative guidelines. But it has clear limits:

1. Lock-in values are empirically calibrated. The Adler equation describes the *form* of coupling correctly (pulling curve, locking threshold), but Δf_{lock} itself cannot be calculated from the Helmholtz model. The nonlinear amplification factor ($\times 9$ at full amplitude) is an estimate from Doc. 0002 Ch. 8, not from an FSI simulation.

2. Zone boundaries are geometry-dependent. The values $f_H/f = 5$ and 3 apply to the reed block calculated in Doc. 0007 (variant C, $\varnothing$ 10 mm opening, $Q_H \approx 8$). With other geometries the boundaries shift.

3. Volume dependence is qualitative, not quantitative. The nonlinear amplification factor scales with A_{eff}^2 , and A_{eff} depends on bellows pressure. At pianissimo a note in Zone 1 may be completely unproblematic — at fortissimo it locks. This dynamics dependence is known in practice but not exactly calculable.

4. Aerodynamic near-field effects are absent. With a shared opening, the airflows of two reeds can influence each other directly (Bernoulli interaction) before chamber resonance comes into play. This effect is not included in the Helmholtz model.

11 Summary

1. Classification: This document is an explanatory model, not a design guide. The calculations provide at best a plausible physical explanation of the observed phenomena. The numerical values (Δf_{lock} , compensation table) are order-of-magnitude estimates — they show the direction, not the destination. Design decisions still require practical tests and experience.

2. The coupling mechanism: Multiple reeds couple through the chamber resonance, not through the weak compliance air spring. Overtones near f_H are resonantly amplified and transfer energy between reeds. At full amplitude, coupling is $\approx 500\times$ stronger than the linear model predicts.

3. The three-zone gradient model:

- Zone 1 ($f_H/f > 5$, low notes): Dense overtones $\rightarrow$ broadband coupling $\rightarrow$ **locking**.
- Zone 2 ($f_H/f \approx 3-5$, middle notes): Individual strong OT $\rightarrow$ selective coupling $\rightarrow$ **ghost peaks**.
- Zone 3 ($f_H/f \approx 2-3$, high notes): Octave/twelfth on $f_H \rightarrow$ massive absorption $\rightarrow$ **response loss**.

Boundaries are gradual. Effects occur near the critical coupling, not exactly at the resonance notes.

4. Remedies:

- **Separate openings** weaken coupling by factor ≈ 7 — the most effective measure.
- **Tune tremolo wider:** $\Delta f_{\text{tuning}} = \sqrt{\Delta f_{\text{target}}^2 + \Delta f_{\text{lock}}^2}$ compensates pulling, but does not help against locking.
- **Reduce chamber depth** (Doc. 0007, variant C) raises f_H and shifts zone boundaries upward — fewer notes are affected.

5. The unified physics: Response (Doc. 0005), tone (Doc. 0008) and tremolo coupling (this document) are three projections of the same chamber impedance $Z_H(f)$. What improves response also improves tone and reduces tremolo coupling. There is no trade-off — the optimum is the same: $f_H/f \geq 3$, minimum volume, large opening.

*Calculation says where to look. Practical testing says what you find.
And experience from a thousand instruments says what matters.*

Dok. 0010

Reed Quality Factor

Reed Quality Factor

Measurement, Material Influence and Clamping

The quality factor Q of the reed determines response and tone (Doc. 0005, 0008). The energy of the vibrating reed leaves the system through two channels: (1) air coupling — the reed as an impulse generator in the gap, and (2) mechanical clamping — bending-zone hysteresis and contact friction at the rivet. Calculations show that the air channel accounts for more than 97% of total losses. The mechanical clamping contributes less than 3%. This explains why almost any combination of materials works — and why the fit of the reed in the slot affects response far more than the plate material. Values are order-of-magnitude estimates.

1 Two Loss Channels

The reed is a mechanical oscillator driven by bellows pressure. The stored vibrational energy leaves the reed through two physically distinct channels:

Channel 1 — Air coupling (dominant): The reed swings through the gap in the reed plate, acting as an impulse generator. With each pass it pushes air through the gap. Losses consist of: (a) acoustic short-circuit through the side gap (pressure equalisation around the reed), (b) squeeze-film damping (viscous forces in the narrow gap), (c) vortex shedding at the gap edges, (d) sound radiation. This channel is material-independent — it depends on gap geometry, not plate material.

Channel 2 — Mechanical clamping (secondary): The bending vibration of the reed creates periodic forces at the clamping point (rivet/screw). In the bending zones — reed at the foot, plate under the rivet, reed block under the plate — a fraction of the elastic energy is converted to heat each cycle through internal friction (hysteresis, $\tan \delta$). Contact friction at the rivet adds further losses. This channel depends on material and clamping quality.

2 Case A — Plucked Reed Without Air Channel

To measure the mechanical channel in isolation, consider a reed that is plucked and allowed to ring down in still air — without channel, without bellows pressure, without flow. Losses: only material hysteresis ($\tan \delta_{\text{steel}} = 0.0002 \rightarrow Q = 5000$) and viscous air friction on the reed surface.

Result: $Q \approx 4000\text{--}4400$. In the instrument: $Q \approx 50\text{--}200$.

Proof: $Q_{\text{plucked}} \approx 4000$ vs. $Q_{\text{instrument}} \approx 100$. The difference (factor 20–80) is the air channel. Practical test: pluck the reed and observe the decay — the relative duration between different reeds shows the differences in Q_{mech} .

Note on τ : The time constant $\tau = Q/(\pi f)$ is a system property. The actually audible ring-down time also depends on the initial energy: $t_{\text{audible}} = \tau \cdot \ln(A_{\text{start}}/A_{\text{threshold}})$. Plucking imposes a deflection x , not a controlled energy. Stored energy $E = \frac{1}{2}kx^2$, where $k \propto EI/L^3$ is the reed spring rate. A heavier reed (stiffer spring) stores more energy for the same deflection $\rightarrow$ louder start $\rightarrow$ audible longer — even with the same Q . With bellows airflow in the instrument this is different: there the bellows pressure times area determines delivered power continuously, not a single energy impulse.

3 Case B — Gap Width and Air Losses

In the second limiting case we assume ideal clamping ($Q_{\text{mech}} = \infty$) and consider only air losses. The side gap s determines two opposing effects:

Narrow gap $\rightarrow$ less short-circuit $\rightarrow$ more drive (good). The side gap has cross-section $A_{\text{gap}} = 2(L+W) \cdot s$ and Poiseuille resistance $R_{\text{gap}} = 12 \mu h_{\text{plate}}/(2(L+W) \cdot s^3)$. Drive efficiency is the pressure divider:

$$\eta = \frac{R_{\text{gap}}}{R_{\text{main}} + R_{\text{gap}}} \quad (1)$$

Narrow gap $\rightarrow$ more squeeze damping $\rightarrow$ more braking (bad). With each pass the reed displaces volume $Q = W \cdot t \cdot v_{\text{tip}}$, which must escape through the side gap. The Poiseuille resistance creates a back-pressure that brakes the reed.

Why high reed plates need narrower gaps: The main opening $A_{\text{main}} = W \times x_{\text{tip}}$ decreases for smaller reeds (bass:

8 mm², very high: 0.5 mm²). With the same side gap, the short-circuit ratio $K = A_{\text{gap}}/A_{\text{main}}$ would be much larger for small reeds — the entire bellows pressure escapes through the side gap. Therefore s must be proportionally smaller.

Limitation: The calculated optima are systematically higher than practical values. The squeeze-film formula overestimates the braking effect, because the reed bends during transit — the gap is narrow only at the foot, wider at the tip. The qualitative statement — narrower is better up to the squeeze limit — remains valid.

4 Mechanical Clamping — Lumped-Mass Model

Plate, reed block and test block are all $\ll \lambda$ (plate 40 mm $\ll \lambda = 8$ m at 440 Hz in brass). No wave propagation in the plate — the transmission coefficient model (plane waves at interfaces) is **incorrect**. Instead: **lumped-mass model** — the plate acts as a concentrated mass.

The impedance of a concentrated mass is $Z = j\omega M$ — purely imaginary, lossless. An ideal mass reflects 100 % of the vibrational energy. Losses arise only from:

1. Bending-zone hysteresis: In the zones where material is periodically deformed (reed at the foot, plate under the rivet, reed block under the plate), a fraction $\tan \delta$ of the elastic energy is converted to heat each cycle. The stiffer and more massive the clamping, the smaller the deformation, the less the loss.

2. Contact friction at rivet/screw: The shear force at the clamping point ($F \approx 0.25$ N for A4 at 1 mm amplitude) causes micro-motion at the contact surface. Rivet (large area, firmly pressed): little motion. Screw (point contact): more micro-slip.

3. Transition zone: Between the rivet and the free bending end there is a zone of ≈ 0.05 – 0.2 mm that is neither rigidly clamped nor freely vibrating. This zone is gradually freed through playing-in. Clamping quality and frequency are coupled through this zone: stiffer termination $\rightarrow$ sharper boundary $\rightarrow$ shorter L_{eff} $\rightarrow$ slightly higher frequency.

Consequence: The more massive and firm the clamping, the higher Q_{mech} . Ranking: test block > vice > reed plate on reed block > hand.

5 Material Comparison

Plate mass determines Q_{mech} : heavy plate $\rightarrow$ higher impedance $\rightarrow$ less deformation $\rightarrow$ less loss. With $Q_{\text{air}} \approx 100$: $Q_{\text{total}} = 1/(1/Q_{\text{air}} + 1/Q_{\text{mech}})$. The difference between brass and magnesium is less than 10 % in Q_{total} — inaudible.

6 The Test Block

The test block (5kg steel) is the most massive termination. Block (10 cm) $\ll \lambda$ (11.6 m at 440 Hz) $\rightarrow$ lumped mass. $\tan \delta_{\text{steel}} = 0.0002 \rightarrow 50\times$ less dissipation than wood. Q_{mech} on the test block is the highest of all configurations.

The tonal difference between test block and instrument comes **not** from the impedance chain, but from the absent chamber (no notch filter, Doc. 0008), the free radiation and the different flow conditions. Pitch remains identical. Response and timbre change because of the chamber, not the clamping.

7 Limits of the Calculation

1. $Q_{\text{air}} \approx 100$ is estimated, not calculated. **2.** The squeeze-film formula overestimates braking (reed bends). **3.** Contact friction at the rivet and reed block coupling are empirical assumptions. **4.** Plate resonances are omitted.

8 Surface Roughness of the Reed Plate Channel

Reed plate slots are now predominantly manufactured by wire EDM (electrical discharge machining). This process produces a characteristic surface with $R_a \approx 0.8\text{--}3.2\text{ }\mu\text{m}$, depending on cutting speed and number of trim cuts. The question: does this roughness affect reed quality?

Roughness from Wire EDM

Process	$R_a\text{ }[\mu\text{m}]$	$k_s \approx 6 \cdot R_a\text{ }[\mu\text{m}]$	Note
Main cut (1 pass)	2.5–3.2	15–19	Fast, economical
+ 1 trim cut	1.2–1.8	7–11	Standard quality
+ 2 trim cuts	0.6–1.0	4–6	Fine, $\pm 0.01\text{ mm}$ tolerance
+ 3 trim cuts (ultra-fine)	0.3–0.5	2–3	Maximum quality, expensive

Roughness vs. Boundary Layer Thickness

In an oscillating flow, a viscous boundary layer of thickness $\delta = \sqrt{2\nu/\omega}$ forms at the wall. If $k_s \ll \delta$: hydraulically smooth (roughness invisible). If $k_s \approx \delta$: transition range. If $k_s \gg \delta$:

hydraulically rough.

For $R_a = 1.6 \mu\text{m}$ (standard EDM, 1 trim cut): $k_s \approx 10 \mu\text{m} = 0.010 \text{ mm}$. Boundary layer thickness δ is 74–309 μm depending on frequency. Ratio $k_s/\delta < 0.15$ at **all** frequencies — surface is hydraulically smooth. Even $R_a = 3.2 \mu\text{m}$ (main cut only) gives $k_s/\delta < 0.3$ — still in the smooth regime.

Diagrams: see PDF (Figs. 6 and 7).

Tolerance vs. Roughness

Squeeze damping depends cubically on the gap ($P_{\text{squeeze}} \propto 1/s^3$). A tolerance deviation of $\pm 0.01 \text{ mm}$ at a nominal gap of 0.03 mm changes squeeze damping by a factor of 3.4. The same roughness ($k_s = 0.010 \text{ mm}$) changes it by less than 0.1 %.

Summary: Surface roughness from wire EDM ($R_a = 0.8\text{--}3.2 \mu\text{m}$) is hydraulically smooth at all frequencies. Finer surfaces ($R_a < 0.8 \mu\text{m}$) provide no acoustic benefit. Dimensional tolerance ($\pm 0.01 \text{ mm}$) determines response, not surface quality. Tolerance is 100× more important than roughness. Economically: 1 trim cut ($R_a \approx 1.6 \mu\text{m}$) with tight tolerance is acoustically optimal.

Aside: Why the Propeller Analogy Does Not Apply

For propellers, golf balls and sharkskin, a rough surface improves performance because roughness triggers boundary-layer turbulence earlier. A turbulent boundary layer has more kinetic energy near the wall, delays flow separation and reduces pressure drag.

This does not apply in the reed plate channel. The propeller effect occurs at $\text{Re} > 10^5$. In the side gap $\text{Re} \approx 100\text{--}200$ — purely laminar flow. With laminar flow, roughness has no effect on flow resistance as long as $k_s \ll s$. The Poiseuille formula $R \propto 1/s^3$ holds unchanged. The channel is straight and parallel — no convex surface, no flow separation, no pressure drag.

The gap edge at the inlet creates local vortices (90° deflection), but the inlet loss is only 2–5 % of the friction loss over the 3 mm channel length — also secondary.

The paradox: Even if roughness increased gap resistance — that would actually be **beneficial** for response, because high gap resistance reduces acoustic short-circuiting (Section 3). But at $\text{Re} = 100\text{--}200$, roughness does nothing.

9 Summary

Classification: Explanatory model. The split into two channels is physically sound. Absolute Q values and gap limits are order-of-magnitude estimates.

1. Two channels: Air coupling $> 97\%$, clamping $< 3\%$. Proof: plucked reed $Q \approx 4000$, in instrument $Q \approx 50\text{--}200$.

2. Gap optimum: Narrower $\rightarrow$ less short-circuit (good), but more squeeze (bad). Small reeds need narrower gaps (smaller A_{main}).

3. Clamping: Lumped-mass. Firmer = better. Rivet $>$ screw. Test block has highest Q_{mech} .

4. Why almost anything works: The air channel dominates and is material-independent. Doubling Q_{mech} changes Q_{total} by $< 3\%$.

5. Ranking: 1. Reed/slot gap (dominant). 2. Chamber geometry (Doc. 0005–0008). 3. Rivet vs. screw. 4. Plate material (minor). 5. Reed block (even less). 6. $\tan \delta_{\text{steel}}$ (negligible).

Closing remark: This document assigns each individual effect its share and classifies most as negligible. This is computationally correct — but it underestimates the sum. Plate material 3 %, reed block 1 %, casing vibration 0.5 %, wax ageing 0.5 %, rivet quality 1 %, plate resonances 1 % — each item individually justifies no attention. But ten such items sum to 10–30 %, and they do not sum linearly: like a butterfly effect, small couplings between the effects can amplify or diminish the result in ways that would not be recognised by examining each contribution alone.

Additionally, this document does not address **ageing**. Wax changes its stiffness over years. Glue joints in the casing — especially with hide glue — gradually fail under the vibrations to which the casing is permanently subjected. Playing-in an instrument is not a myth: the transition zone at the clamping point is gradually freed (Section 4), glue joints settle, and the wood of the reed block and casing changes its damping properties under cyclic loading. An absolutely vibration-free casing does not exist — it is part of the system.

The margin between a good and an excellent instrument is small — but perceptible. To say that the reed block or the vibration behaviour of the casing has no noticeable influence is

probably an exaggeration. The calculation shows the ranking. The sum of the small things that no model captures individually makes the difference that the instrument maker hears and the player feels.

This document provides orders of magnitude to assess the relevant factors. It does not replace the practice of the voicer and the trained ear. Comparing reed plates by listening reveals all the subtleties that would be practically impossible to capture metrologically with realistic effort. Experience and ear training are the key to success.

The reed gives its energy to the air — not to the plate. The plate holds the reed. The air makes the sound.

Dok. 0011

Channel Geometry of the Reed Plate

Channel Geometry of the Reed Plate

Corner Shapes, Taper and Acoustic Effect

The slot in the reed plate through which the reed vibrates is not merely a geometric frame — through its gap geometry it governs the energy supply, limits the minimum gap width via torsional vibration, and modifies the acoustic short-circuit when tapered. Plate thicknesses range from 2 mm (C6) to 15 mm (50 Hz). At full volume the reed swings at least twice as far as the plate thickness — energy input occurs only during the transit through the channel.

1 The Reed Swings Beyond the Channel

The reed tongue deflects and swings through the slot in the reed plate. At full volume the tip amplitude x_{tip} is at least 1.2–1.5 times the plate thickness h . Total stroke: $2 \times x_{\text{tip}} \geq 2h$.

Energy input from bellows pressure occurs **only** while the reed is inside the channel. Above and below the channel there is no pressure difference — the reed swings freely.

Diagrams: see PDF (7 figures).

2 Duty Cycle — Time Fraction Inside the Channel

$$\text{Duty} = \frac{2}{\pi} \cdot \arcsin\left(\frac{h}{2x_{\text{tip}}}\right) \quad (1)$$

For $x_{\text{tip}} = 1.3 \times h$ the duty cycle is $\approx 25\%$. The reed spends only one quarter of the period inside the channel. It passes through near the zero crossing, where velocity is **maximum** ($v = \omega \cdot x_{\text{tip}}$). Mean velocity during channel transit is $\approx 1.5\times$ higher than over the full cycle.

For comparison of different channel geometries (parallel vs. tapered), only the channel region counts. Since both have the same duty cycle, relative comparison is valid.

3 Torsional Vibration — the Minimum Gap

In addition to the fundamental bending mode, the reed has a torsional mode ($f_T = 5\text{--}14 \times f_{\text{bending}}$). During transient excitation, torsion is briefly excited. Lateral deflection at $\theta = 1^\circ$:

$y_{\text{lateral}} = \theta \cdot W/2 = 0.04\text{--}0.10\text{ mm}$ — the same order as the side gap.

Consequence: The minimum gap is limited by the torsional amplitude during transient excitation, not by manufacturing tolerance alone. Bass: $s_{\text{min}} \approx 0.10\text{--}0.15\text{ mm}$. High treble: $s_{\text{min}} \approx 0.05\text{ mm}$.

4 Tapered Channel Widening

To give torsion room without increasing the top gap: $s(z) = s_0 + (s_1 - s_0) \cdot z/h$. Cost: gap resistance decreases, drive efficiency decreases.

The taper effect scales with the **taper angle**, not with the absolute widening:

- Bass ($h = 15\text{ mm}$): 0.04 mm widening $\rightarrow$ angle $0.15^\circ \rightarrow -3\%$ drive (negligible)
- C6 ($h = 2\text{ mm}$): 0.04 mm widening $\rightarrow$ angle $1.15^\circ \rightarrow -5\%$ drive (noticeable)

Key finding: Tapered widening is more critical for thin plates (treble/high) than in the bass. The same absolute widening produces an $8\times$ steeper angle at 2 mm plate thickness than at 15 mm .

5 Drive Area — Why Thick Plates Are Needed

Drive force $F = \Delta p \times W \times h$. Bass 50 Hz : $F = 150\text{ mN}$ ($10 \times 15\text{ mm}$). C6: $F = 7\text{ mN}$ ($3.5 \times 2\text{ mm}$). Factor of $21\times$. This compensates for the heavier bass reed.

6 Why Overly Thick Plates Harm High Notes

Experience shows that reed plates for very high notes must not be too thick. A 4 mm plate for C6 performs worse than a 2 mm plate. The explanation:

Effect 1 — Drive saturates: $\eta = R_{\text{gap}}/(R_{\text{main}} + R_{\text{gap}})$ increases with h but approaches 100% asymptotically. For C6 ($s = 0.025\text{ mm}$), $\eta > 90\%$ already at $h \approx 1\text{ mm}$. More thickness yields almost no additional drive.

Effect 2 — Squeeze brake grows quadratically: $R_{\text{sq}} \propto h$ (longer channel), braking area $\propto h$. Result: $P_{\text{squeeze}} \propto h^2$. The tighter the gap, the steeper the increase ($\propto 1/s^3$). High notes with narrow gaps are particularly sensitive.

Effect 3 — Air mass inertia: Air column in slot has $m = \rho \cdot W \cdot s \cdot h$. Acceleration force $F = m \cdot \omega^2 \cdot x_{\text{tip}}$. This term grows with $h \times \omega^2$. At 50 Hz: $\omega^2 \approx 10^5$. At 1000 Hz: $\omega^2 \approx 4 \times 10^7$ — 400× larger.

Why practice agrees: For C6, $\eta > 90\%$ at 1 mm — more thickness gives almost no drive, but the squeeze brake keeps growing quadratically. For bass, $\eta > 90\%$ only at ≈ 5 mm. Additionally, $R_{\text{squeeze}} \propto 1/s^3$ is 14× smaller for bass ($s = 0.06$ mm) than C6 ($s = 0.025$ mm).

Limitation: The squeeze model overestimates the braking effect (Doc. 0010). The relative statement — high notes are more sensitive to overly thick plates — is physically sound and consistent with experience.

Diagrams: see PDF (Figs. 8-10).

7 Summary

- 1. Duty cycle $\approx 25\%$:** The reed is inside the channel only one quarter of the time — at maximum velocity (zero crossing).
- 2. Torsion determines minimum gap:** Lateral deflection 0.04–0.10 mm during transient excitation.
- 3. Tapered widening:** Benefit: room for torsion. Cost: scales with taper angle — negligible for bass, significant for C6.
- 4. Drive area:** $F \propto W \times h$. Thick bass plates deliver 21× more force than C6.
- 5. Too thick plates harm high notes:** Drive saturates ($\eta \rightarrow 100\%$), but squeeze brake grows quadratically ($\propto h^2$) and air inertia grows with $h \times \omega^2$. Practical thicknesses (2 mm at C6, 15 mm at bass) are determined by this balance.

The channel is not just a slot — it is the place where pressure becomes motion.

Dok. 0012

Reed Stiffness

Reed Stiffness

Calculation, Material Selection and Effect on Frequency

The stiffness of the reed determines two properties that are in conflict: response and tuning stability. A soft reed responds more easily but detunes more strongly with bellows pressure. A stiff reed holds the pitch more steadily but requires more pressure to start vibrating.

1 Stiffness and Frequency

Frequency and stiffness both depend on geometry, but to **different degrees**:

$$f \propto \frac{t}{L^2}, \quad k = \frac{E \cdot W \cdot t^3}{4 \cdot L^3} \quad (1)$$

Stiffness k depends **cubically** on t , while frequency depends only linearly. 10 % thinner = 27 % softer, but only 10 % flatter.

2 Detuning by Bellows Pressure

The Bernoulli effect creates a “negative spring”: $k_{\text{eff}} = k_{\text{reed}} - k_{\text{Bernoulli}}(p)$. The frequency shift: $\Delta f/f \approx -k_{\text{Bernoulli}}/(2 \cdot k_{\text{reed}})$.

Result: Bass ($k \approx 80\text{--}150\text{ N/m}$): 4–11 cents/kPa (audible). Treble ($k \approx 250\text{--}400\text{ N/m}$): 0.3–0.8 cents/kPa (barely audible).

3 The Trade-off

$\Delta f \times p_{\text{threshold}} \approx \text{const}$ — good response and stable tuning are physically incompatible.

4 Tuning Weight vs. Thinning

Adding mass: Lowers f , k unchanged $\rightarrow$ same detuning.

Thinning: Lowers f and k ($k \propto t^3$) $\rightarrow$ 37 % more detuning per 10 % thinning.

5 Summary

1. $k \propto t^3$: cubic coupling.
2. Detuning $\propto 1/k$.
3. $\Delta f \times p_{\text{th}} \approx \text{const}$.
4. Tuning weight is more pitch-stable than thinning.
5. Profiling decouples mass and stiffness.

*The reed seeks its equilibrium between freedom and control — just
like the player.*

Dok. 0015

Acoustic Coupling

and Impedance Matching

Acoustic Coupling and Impedance Matching

Channel Geometry, Slot Shape and Reed Block Design

Note: This document presents an explanatory model. The numerical values given are order-of-magnitude estimates — they provide no constructive specifications. The optimum is heard, not calculated.

Analysis of acoustic coupling between reed and chamber as a two-port system. Transition geometry, constriction as acoustic mass, coupling centroid and impedance mismatch. Reference: Doc. 0005-0013.

1 The Chamber as a Two-Port System

In the previous documents (Doc. 0004-0008), the chamber was treated as a Helmholtz resonator with a single opening (valve). Doc. 0013 showed that the reed never completely closes the channel — the side gap always remains open. The chamber therefore has **two ports**:

Port 1 (reed end): The reed pulsates and creates pressure impulses. The effective area fluctuates between the side gap ($A_{\min} \approx 10 \text{ mm}^2$) and the slot area ($A_{\max} \approx 26 \text{ mm}^2$).

Port 2 (valve): Stationary opening with $A_2 \approx 300 \text{ mm}^2$. Sound radiates from here.

The conductance $G = A/l$ determines how easily air flows through each port. Port 1 (reed) carries time-averaged $\approx 70\%$ of total conductance — the reed is the primary port, not the valve.

$$f_H = \frac{c}{2\pi} \sqrt{\frac{S_1/l_1 + S_2/l_2}{V}} \quad (1)$$

Consequence: All calculations in Doc. 0004–0008 that derive f_H only from the valve opening must be understood as lower bounds. The actual chamber resonance lies higher.

2 Is the Fold Acoustically Visible?

For a bass at 50 Hz, the wavelength is $\lambda = c/f = 6860$ mm. The largest chamber dimension (chamber length 120 mm) is only 1.7 % of λ . A partition wall (30 mm) is 0.44 % of λ .

When $d/\lambda < 0.1$, the wave diffracts completely around the obstacle — it does not exist acoustically. When $d/\lambda > 0.5$ it acts as a reflector.

At 50 Hz, **all** chamber geometries lie deep in the regime $d \ll \lambda$. Only above ≈ 1400 Hz (28th overtone) does the partition wall begin to approach the limit $d/\lambda = 0.1$.

The analogy with an optical waveguide confirms this: a smooth, round tube with constant cross-section has no reflection coefficient — the wave follows the curve as long as the bend radius is large compared to the wavelength.

Coltman correction: Coltman compensates not the bend itself, but the cross-section change and path-length difference at the fold. $Z = \rho c/A$ remains constant at constant cross-section. The path-length correction scales with d^2/r — at $d/\lambda < 0.01$ it is negligible.

Conclusion: In the bass chamber, the fold plays no role for the fundamental and the first overtones. Matching occurs at the openings (Port 1 and Port 2), not at the fold.

3 Constriction = Acoustic Mass

Every cross-section reduction in the channel acts as a concentrated acoustic mass: $m = \rho \times l/A$. This holds regardless of whether the constriction is in a straight tube, at a fold, or at the slot→chamber transition.

The Coltman compensation in folded organ pipes is fundamentally a constriction at the deflection point. The mechanism changes (waveguide vs. lumped element), but the physical element (constriction = mass) remains the same.

4 Pressure Build-up Before the Constriction

The reed's pressure impulse backs up before the constriction at the slot→chamber transition. Three scenarios:

Too wide: Pressure escapes immediately into the volume — weak back-reaction, weak coupling.

Optimal: Pressure briefly backs up — maximum back-reaction, best coupling.

Too narrow: The impulse is predominantly reflected — energy does not enter the volume.

5 Impedance Matching: Transition Geometry as Transformer

The observation that different transition geometries change **response** but not **filter action** can be modelled cleanly as a matching problem:

Source: Bellows pressure p_{bellows} with internal resistance R_i depending on the system's flow resistance. $R_i \propto 1/A^2$ (Bernoulli).

Load: The reed with mechanical impedance R_L (fixed, determined by stiffness and mass, Doc. 0012).

Transformer: The transition geometry at the reed end determines how efficiently flow energy is converted to reed energy.

$$P/P_{\max} = \frac{4 R_L R_i}{(R_i + R_L)^2} \quad (2)$$

Maximum at $R_i = R_L$ — classical matching condition. Too narrow: $R_i \gg R_L \rightarrow$ blockage. Too wide: $R_i \ll R_L \rightarrow$ short circuit. Optimal: $R_i = R_L \rightarrow$ maximum power transfer bellows→reed.

Crucial: The filter action (f_H , notch filter) depends on the resonator — chamber volume and opening area. The transformer (transition geometry) changes matching efficiency, not resonance. This is exactly what is observed: different response, same timbre.

6 Equivalent Circuit: Two Coupled Resonators

The separation of response and filter action can be represented exactly as an electroacoustic analogy. Reed and chamber are

each a series resonator (RLC), connected by a coupling element:

Circuit 1 (reed): Mass $m \rightarrow$ inductance L_1 . Stiffness $k \rightarrow$ capacitance $C_1 = 1/k$. Damping $\rightarrow$ resistance R_1 . $f_1 = 1/(2\pi\sqrt{L_1C_1}) \approx 50$ Hz.

Circuit 2 (chamber): Acoustic mass $\rho l/A \rightarrow L_2$. Volume compliance $V/(\rho c^2) \rightarrow C_2$. $f_H = 1/(2\pi\sqrt{L_2C_2})$.

Coupling: Mutual impedance $M = k \times \sqrt{L_1L_2}$. The coupling coefficient k (0–1) is determined by the transition geometry.

The transfer function $H(f) = I_2/V_1$ shows two resonance peaks (at f_1 and f_H). Crucially: k changes the **height** of the peaks (energy transfer = response), but not their **position** (frequency = filter action). Conversely, C_2 (chamber volume) shifts the f_H peak without changing response.

Calculable: With the physical values (mass, stiffness, volume, opening area) the transfer function is quantitatively determined. k determines response, C_2 determines filter action — these parameters are independent.

7 Valve Position — Fine Adjustment

In practice the valve opening can be offset by a few millimetres to at most one centimetre from the chamber end. The shift changes the effective acoustic neck length and thus f_H .

In the practical range (0–15 mm), f_H shifts by a few percent. Valve position is a fine adjustment — the main lever is at the reed end (Port 1), where 70 % of conductance resides.

8 Limits of the Model

The two-port model shows relationships and directions. It does not say where the optimum lies.

(a) Nonlinearity: The reed switches periodically between small and large opening. The actual impedance modulation (Doc. 0013) is highly nonlinear.

(b) Tonal target: The “optimal” spectrum depends on the desired tonal character. The calculation shows the map; the ear walks the path.

(c) Interactions: Gap width (Doc. 0010), stiffness (Doc. 0012), chamber volume, valve opening and transition geometry all act together.

This document is an explanatory model, not a design guide. Design decisions still require practical tests and experience.

9 Where Does the Reed Couple? — Centroid of Volume Flow

The assumption that coupling occurs at the absolute end of the reed (tip) is a simplification. The reed is a cantilever — its vibration shape follows the first Euler-Bernoulli bending mode.

Volume flow per unit length $dQ/dx \propto y(x)$ increases steeply towards the tip. The weighted **centroid of volume flow** lies at $x/L = 0.73$ — not at the tip ($x/L = 1.0$). 50 % of total volume flow arises in the last 23 % of the reed length.

For a bass reed with $L = 70$ mm the centroid lies at 51 mm from the clamping point — about 19 mm before the tip.

Impedance Mismatch

The mechanical impedance of the reed and the acoustic impedance of the air differ by orders of magnitude. The reed is a **current generator** (high impedance) — it drives a volume flow that the air accommodates.

Z_{reed} lies a factor 50–500 above Z_{air} . The coupling is **not matched** in the sense of maximum power transfer ($Z_{\text{source}} = Z_{\text{load}}$). This is physically correct: the reed is self-excited and vibrates at its natural frequency, not at the chamber resonance. The air is “carried along”, not “driven”.

This is consistent with Doc. 0010: over 97 % of energy output occurs through air coupling, less than 3 % through the mechanical clamping.

Consequence: Position of the Plate Edge

When the reed plate covers the full reed length, the entire slot couples into the chamber. If the reed protrudes beyond the plate edge (deflection), the most energetic part — the last 20–30 % of the reed — bypasses the chamber and radiates directly into free space.

The position of the plate edge relative to the reed therefore determines what fraction of the volume flow is coupled into the chamber and what fraction radiates directly.

10 Summary

1. **Two-port system:** The chamber is open at both ends. Port 1 (reed) carries $\approx 70\%$ of conductance. The single-port model underestimates f_H .
2. **Fold invisible:** At 50 Hz, the largest chamber dimension is 1.7% of λ . Chamber shape is irrelevant — only volume counts.
3. **Impedance matching:** The transition geometry acts as a transformer between bellows pressure (source) and reed (load). Maximum power transfer at $R_i = R_L$. Filter action (f_H) depends on the resonator, not the transformer.
4. **Equivalent circuit:** Reed and chamber form two coupled RLC resonators. k determines response (energy transfer), $C_2 = V/(\rho c^2)$ determines filter action (f_H). These parameters are independent.
5. **Valve position = fine adjustment:** A few percent f_H shift. The main lever is at the reed end.
6. **Centroid at $x/L = 0.73$:** Coupling does not occur at the reed tip, but $\approx 27\%$ before it. Plate edge position determines the coupling.
7. $Z_{\text{reed}} \gg Z_{\text{air}}$: The reed is a current generator. Impedances are not matched — the air is carried along, not driven.
8. **Optimum is heard:** The calculation shows relationships and directions. Experience and ear training remain the key to success.

*The reed feeds, the chamber shapes, the valve radiates.
The geometry at the transition determines the coupling.*

Dok. 0016

Overtone Modes of the Bass Reed

Profiling and Inharmonicity

Overtone Modes of the Bass Reed

Profiling, Inharmonicity and Tonal Character

Note: This document presents an explanatory model. The FEM calculation captures the bending modes of an idealised beam — flow coupling, nonlinear amplitude and gap effects are not included.

*Calculation scripts: The complete FEM calculations with all tables are documented in two separate Python scripts deposited in the same directory on GitHub: **pruefskript_0016_obertonmoden.py** (profiles, weights, stiffness, node positions) and **pruefskript_0016_chromatisch.py** (chromatic analysis over 2 octaves, $f_2/f_1 = \text{const}$ for uniform profile, absolute f_2 vs. f_H ranges) and **pruefskript_0016_praxis.py** (practical profiling with migrating thinnest point, thinning depth, optimum search).*

FEM calculation of the bending modes of a bass reed ($L = 70 \text{ mm}$) with six different thickness profiles. Mode shapes, frequency ratios, inharmonicity and consequences for tone and chamber coupling. Reference: Doc. 0005-0009, 0012, 0015.

1 Overtones of a Cantilever Are Not Harmonic

The reed is a cantilever beam, clamped at one end. The natural frequencies of the bending modes do not follow the harmonic series ($1 : 2 : 3 : 4 \dots$), but the cantilever series:

$$f_n/f_1 = (\beta_n/\beta_1)^2 \quad (1)$$

with $\beta_1 L = 1.8751$, $\beta_2 L = 4.6941$, $\beta_3 L = 7.8548$, $\beta_4 L = 10.9955$. This gives ratios $1 : 6.27 : 17.55 : 34.39$ — the 2nd overtone lies not at the octave ($2\times$), but at more than two and a half octaves ($6.27\times$).

The harmonics in the sound do not come from these bending modes. They arise from the periodic interruption of the airflow — the reed acts as an impulse generator (Doc. 0013, 0015). The pulse shape produces harmonic overtones ($1f, 2f, 3f \dots$), even though the mechanical modes lie entirely elsewhere.

Why are the mechanical modes important nonetheless? Because they determine at which frequencies the reed **can be resonantly excited** — by the chamber (Doc. 0009), by neighbouring reeds, or by transient pressure fluctuations during playing.

2 Six Thickness Profiles

Profiling changes the mass distribution and the local bending stiffness. Six profiles are compared, all with $t_0 = 0.40\text{ mm}$ at the foot.

Profile	t_{tip}	Thinning	f_1	f_2/f_1	f_3/f_1
Uniform	0.40 mm	0 %	66.8 Hz	6.27	17.55
Linear 80 %	0.32 mm	20 %	68.5 Hz	5.71	15.57
Linear 50 %	0.20 mm	50 %	72.6 Hz	4.79	12.36
Linear 30 %	0.12 mm	70 %	77.5 Hz	4.07	9.94
Parabolic 50 %	0.27 mm	50 %	77.9 Hz	5.01	12.89
Inv. parabolic	0.20 mm	50 %	61.8 Hz	4.97	13.00

3 Overtone Ratios Compared

No profile approaches the harmonic series. Even at 70 % thinning, Mode 2 lies at $4.07\times$ instead of $2\times$. The inharmonicity is a fundamental property of the cantilever — only a string oscillator would have harmonic overtones.

4 Tuning Weights: The Opposite Effect

Profiling reduces f_2/f_1 — the modes move closer together. Tuning weights at the tip act **in the opposite direction**: they lower f_1 strongly, but higher modes less so. f_2/f_1 **increases** with added weight.

Weight	Mass	% m_Z	f_1	f_2/f_1	f_3/f_1
None	0 g	0 %	66.8 Hz	6.27	17.55
0.10 g	0.10 g	5.7 %	60.2 Hz	6.37	18.04
0.20 g	0.20 g	11.4 %	55.2 Hz	6.58	18.96
0.40 g	0.40 g	22.9 %	48.1 Hz	7.11	21.03
1.50 g	1.50 g	85.9 %	31.4 Hz	9.89	30.84

The reason: The weight sits at the vibration maximum of Mode 1. Mode 1 is maximally slowed. Mode 2 has less amplitude at the tip → is less slowed → the ratio increases.

5 Combination: Profiling + Weight

In practice both are combined. Thinning reduces f_2/f_1 , weight raises it again. Linear 30 % alone: $f_2/f_1 = 4.07$ at $f_1 = 77.5$ Hz. Linear 30 % + 0.20 g: $f_2/f_1 = 4.64$ at $f_1 = 50.2$ Hz.

6 Same Target f_1 , Different Paths

At $f_1 = 50$ Hz: uniform + weight gives $f_2/f_1 \approx 6.7$. Linear 50 % + weight: ≈ 5.3 . Linear 30 % + weight: ≈ 4.6 . Profiling wins — even with weight.

Response: The profiled reed has lower stiffness at the same f_1 → better response (Doc. 0012). Profiling + less weight is better than weight alone.

7 Mode Shapes and Node Positions

Profiling shifts the vibration nodes towards the tip. For the uniform reed the first node of Mode 2 is at $x \approx 55$ %. With Linear 30 % it migrates to $x \approx 65$ %. In heavily profiled reeds the vibrational energy concentrates at the light end.

8 Inharmonicity in Cents

All overtones lie thousands of cents above the harmonic series. Profiling reduces the deviation by 25–35 %.

Consequence: Bending modes never coincide with harmonic overtones of the fundamental. Chamber coupling (Doc. 0009) affects the mechanical modes — whether they lie near f_H .

9 Stiffness and Fundamental Frequency

Profiling has two simultaneous effects: f_1 increases (less mass), k decreases (thinner cross-section). A profiled reed for the same pitch must be longer or thicker at the foot — and at the same frequency has lower stiffness, hence better response (Doc. 0012).

This is the real practical benefit: Lower stiffness at the same fundamental frequency → better response.

10 Chamber Coupling: Where Do the Modes Fall?

Mode 2 for the uniform reed lies at ≈ 420 Hz — right in the typical f_H range. Profiling shifts Mode 2 downward and can push it deliberately out of the f_H range.

11 Practical Profiling: Position of the Thinnest Point

In practice the reed is thickest at the rivet and tapers smoothly. The thinnest point can be at the tip or between tip and mid-span — the transition is never abrupt.

Surprising result: The position of the thinnest point has little influence on f_2/f_1 . Between $x_{\min} = 0.7L$ and $1.0L$ the plateau is practically flat. The **depth** of the thinning determines f_2/f_1 ; the position determines f_1 .

12 Optimal Thinning Depth

f_2/f_1 has a **minimum at $\approx 70\%$ thinning** ($t_{\min} \approx 0.12$ mm at $t_0 = 0.40$ mm). Beyond this, f_2/f_1 rises again.

Thinning	$t_{\min}$	f_1	f_2/f_1	f_3/f_1
0 %	0.40 mm	66.8 Hz	6.27	17.55
40 %	0.24 mm	68.4 Hz	4.70	12.61
50 %	0.20 mm	67.7 Hz	4.35	11.50
60 %	0.16 mm	65.4 Hz	4.07	10.50
70 %	0.12 mm	60.4 Hz	3.94	9.75
80 %	0.08 mm	49.8 Hz	4.25	9.61

13 Chromatic Analysis: 2 Octaves, Uniform Scale

For a uniform reed, $f_2/f_1 = 6.27 = \textbf{constant}$, independent of thickness. Since $f \propto t$, all modes scale identically — the ratio cancels. The scale fixes f_2/f_1 for all notes.

The lower octave (A1-A2) is the **critical zone**: f_2 lies at 345–689 Hz — right in the typical f_H range. The upper octave: $f_2 > 700$ Hz, far above f_H . This explains why ghost tones occur mainly in the low notes.

14 Summary

1. **Cantilever overtones are inharmonic:** 1 : 6.27 : 17.55 : 34.39. The audible harmonics come from the impulse generator, not the bending modes.
2. **Profiling compresses the ratios:** Optimum at $\approx 70\%$ thinning ($f_2/f_1 \approx 3.94$). Beyond this f_2/f_1 rises again.
3. **Position of thinnest point:** Between 70 % and 100 % of length, little difference. Depth determines f_2/f_1 ; position determines f_1 .
4. **Tuning weights widen the ratios:** Opposite effect to thinning. Profiling + less weight is better than weight alone.
5. $f_2/f_1 = \textbf{const for uniform profile}$: The scale fixes the ratio for all chromatic notes.
6. **Critical zone = lower octave:** f_2 for A1-A2 lies at 345–689 Hz — right in the f_H range. Ghost tones occur there.
7. **Primary benefit:** Lower stiffness at the same $f_1 \rightarrow$ better response. Shifting Mode 2 out of the f_H range.

The reed vibrates inharmonically. The sound is harmonic.

Profiling changes the response — the overtones are made by the air.

Dok. 0017

Treble Reeds

Overtone Modes F3 to C6

Treble Reeds

Overtone Modes F3 to C6

Note: This document presents an explanatory model. The interpolated scale is an estimate — real reed plates vary depending on the manufacturer.

*Calculation script: The complete FEM calculation with all tables is deposited in the same directory on GitHub: **pruef-skript_0017_diskant.py** (W independence, uniform minimum thickness, calibrated profiling, f_2 vs. f_H , bass/treble comparison). FEM calculation of the bending modes of treble reeds with variable scale ($L = 37 \rightarrow 14$ mm, $W = 3.7 \rightarrow 1.7$ mm). Calibrated thickness for each semitone. Comparison with the bass reeds from Doc. 0016.*

1 Width W Has No Effect on Pitch

For an Euler–Bernoulli cantilever:

$$f = \frac{\beta^2}{2\pi L^2} \times t \times \sqrt{\frac{E}{12\rho}} \quad (1)$$

Width W cancels out: $I = Wt^3/12$, $A = Wt$, so $I/A = t^2/12$. Pitch is determined solely by t and L . W influences mass ($\propto W$), volume flow ($\propto W$), loudness and torsional stiffness ($\propto W^3$) — but not the pitch.

W determines loudness and response, not pitch.
Hence W and L decrease steadily from low to high — the low notes require more volume flow.

2 Treble Scale: L and W Decrease Steadily, t is Profiled

Unlike the bass, in the treble both length and width vary continuously. The foot thickness (t_0) is always slightly greater than the theoretical minimum thickness — profiling (thinning towards the tip) adjusts the pitch.

Note	f	L	W	t_0	t_{tip}	Thin.	f_2/f_1	f_2
F3	175	37.0	3.70	0.298	0.223	28 %	5.32	348
C3	131	29.0	3.05	0.133	0.104	24 %	5.33	697
C4	262	22.8	2.51	0.161	0.132	20 %	5.45	1426
C5	523	17.9	2.06	0.197	0.169	16 %	5.63	2947
C6	1047	14.0	1.70	0.244	0.217	12 %	5.82	6094

$f_2/f_1 = 5.3\text{--}5.8$ (profiled) — lower than the 6.27 of the uniform reed, because profiling compresses the modes. The light profiling in the treble (12–28 %) lowers the ratio less than in the bass (30–50 %).

3 Absolute f_2 : Where Does Mode 2 Hit the Chamber Resonance?

In the bass the critical zone is A1–A2 ($f_2 = 345\text{--}689$ Hz, f_H bass = 200–700 Hz). In the treble everything shifts upward: f_2 lies for C2–C3 at 410–820 Hz — **below** the treble chambers. Only from C4 ($f_2 \approx 1640$ Hz) does f_2 begin to hit the treble chamber range. The critical zone in the treble is **C4–C5** ($f_2 = 1640\text{--}3280$ Hz).

4 Profiling in the Treble: Less Room for Adjustment

Treble reeds are lightly profiled: 12–25 % thinning (bass reeds: 30–50 %). The thinnest point is at the tip for the higher notes, and somewhat towards the middle for the lower ones. Profiling reduces f_2/f_1 by 0.4–0.9 (from 6.27 to 5.4–5.8). In the bass the reduction was 1.0–2.3. Less room for adjustment with thinner reeds.

5 Stiffness and Mass

Stiffness increases exponentially: factor 100 from C2 (4.8 N/m) to C6 (489 N/m). Mass decreases from 117 mg to 47 mg. High notes: stiffer, lighter, higher threshold pressure, faster onset.

6 Comparison: Bass vs. Treble

Common: $f_2/f_1 = 6.27 = \text{constant}$. The cantilever physics are identical.

Difference 1 — Scale: In the bass, one reed length covers 2 octaves. In the treble, L , W and t all vary simultaneously.

Difference 2 — Profiling: Bass: 30-50 %, effect -1.0 to -2.3 . Treble: 12-25 %, effect -0.4 to -0.9 .

Difference 3 — Critical zone: Bass: A1-A2 (f_2 in the f_H range of the bass chambers). Treble: C4-C5 (f_2 in the f_H range of the treble chambers). The lower treble notes (F3-B3) are non-critical.

7 Summary

1. **W has no effect on pitch.** $f \propto t/L^2$. W determines loudness and response.
2. $f_2/f_1 = 6.27$ **(uniform) = identical to bass.** Profiled: 5.3-5.8 depending on thinning depth.
3. **Actual foot thickness > minimum thickness.** Profiling raises f_1 — the tuner starts thicker and grinds down.
4. **Critical zone: C4-C5.** $f_2 = 1400\text{-}3000\text{ Hz}$ hits the treble chambers. Lower notes (F3-B3) non-critical.
5. **Stiffness:** Factor 100 over 4 octaves.

Bass and treble obey the same physics. The scale changes the absolute frequencies — the ratios remain.

Dok. 0018

Audibility of Bending Modes

Chamber Notch, Transients, Torsion

Audibility of Bending Modes

Chamber Notch Filter, Transients and Torsion

Correction to Doc. 0016: Bending modes can become audible under certain conditions. Three mechanisms: chamber resonance, transient onset, torsion. Reference: Doc. 0009, 0011, 0015-0017.

*Calculation script: **pruefskript_0018_hoerbarkeit.py** (modal participation, chamber notch filter, transients, torsional modes) in the same directory on GitHub.*

1 Correction: Bending Modes Are Not Always Inaudible

Doc. 0016 showed that the inharmonic bending modes are normally 30–50 dB quieter than the harmonic overtones. In practice however: with a tuneable resonance chamber, Mode 2 can be excited in some reeds so strongly that **only Mode 2 is audible**. And during the transient onset, Mode 2 shapes the tonal character.

The fundamental statement stands: Harmonic overtones dominate in the steady state. But Mode 2 is not negligible.

2 Chamber as Notch Filter: Mode 2 Becomes Audible by Subtraction

The chamber acts as a **notch filter** (band-rejection, Doc. 0005-0008) — it **suppresses** frequencies at f_H . The chamber does not amplify Mode 2. Mode 2 becomes audible because the fundamental and its harmonics are attenuated, while Mode 2 lies **outside the notch**.

Conditions: (a) f_H hits a harmonic of $f_1 \rightarrow$ fundamental is attenuated. (b) f_2 (bending mode) does **not** fall in the notch $\rightarrow$ Mode 2 survives. This is subtraction, not addition.

Why does this not occur in every reed? Profiling determines f_2 . For a uniform reed, $f_2 \approx 6.27 \times f_1$ — close to $6 \times f_1$ (a harmonic). With stronger profiling ($f_2 \approx 4\text{--}5 \times f_1$), Mode 2 falls between the harmonics and survives the notch.

Cross-references: Doc. 0016 · Doc. 0017

In the accordion this affects notes with response problems: Chamber coupling is strongest there, reed plates are more heavily profiled, and the interplay of profiling and chamber geometry happens to satisfy the condition $f_H \neq f_2$.

3 Transient Excitation: The Onset

When a key is pressed, the bellows pressure arrives as a step function. Every bending mode is initially excited. Mode 2 starts at ≈ -25 dB and decays in ≈ 40 ms.

Mode 2 is audible during the transient — as a brief tonal colouring in the first 50–100 ms. Different reeds have different $f_2 \rightarrow$ different tonal character.

4 Torsional Vibrations

Torsional modes lie above the bending modes: $f_{T1} \approx 14\text{--}22 \times f_1$. Torsion itself is barely audible as a pitch — but lateral deflection leads to channel contact in bass reeds.

At 1° torsion: bass reed ($W=8$ mm): $\Delta y = 0.07$ mm $>$ gap 0.04 mm $\rightarrow$ contact possible. **Scraping should not occur in normal operation** — the gaps are large enough. But in **cold instruments** scraping is normal: Metal contracts ($\approx 2\text{--}3$ μ m at $\Delta T = -17$ K), the gap narrows, tolerance decreases. As the instrument warms up, the scraping disappears.

Paradox: The most precisely fitted reed plates — the most expensive — are most susceptible to scraping in the cold. Narrow gaps (better response) mean less tolerance for torsion. A plate with 25 μ m gap loses 12 % headroom from a 3 μ m contraction — one with 50 μ m only 6 %. The cheap plate with a large gap never scrapes; the expensive one with the perfect gap scrapes in winter.

5 Summary

1. **Steady state, no chamber resonance hit:** Mode 2 at -37 dB — not audible. Harmonics dominate.
2. **Chamber as notch filter:** The chamber attenuates harmonics. If f_2 lies outside the notch, Mode 2 survives —

Cross-references: Doc. 0016 · Doc. 0017

audible by subtraction. Particularly affects notes with response problems.

3. **Transient:** Mode 2 at -25 dB, duration ≈ 40 ms. Shapes tonal character.
4. **Torsion:** $f_{T1} \approx 14-22 \times f_1$. Channel contact not expected in normal operation. Normal in **cold instruments** — metal contracts, gap narrows. Disappears as the instrument warms up.
5. **Correction:** “Bending modes inaudible” applies only in steady state without a chamber resonance hit.

Bending modes are quiet — but not silent. The chamber can wake them, the transient lets them briefly sound, torsion makes them audible as scraping.

Dok. 0500

Aesthetic Forensics

of Accordion Cases

Guide to Aesthetic Forensics

of Accordion Casework

Note on Methodology

This guide focuses exclusively on external design features (grille, veneer, fittings). Note that the technical classification (reed plates, mechanics, reed blocks) forms a separate chapter and must be considered independently of the outer casing. A historic casework may well contain later technical components.

1. Art Nouveau Aesthetics (ca. 1895-1914)

Art Nouveau is not mere imitation of nature, but an abstraction of it. Its aim was to ennoble everyday objects through organic forms.

- **Line quality:** Dominance of the so-called “whiplash line”. Forms appear fluid, undulating and asymmetric. Hard edges and mathematical grids are rare.
- **Motifs:** Floral inspiration: tendrils, blossoms and insects. Even when patterns are arranged symmetrically, they appear alive and “grown”.
- **Grille character:** The grille functions as a decorative element that breaks the surface organically. Decoration is flat and “breathing”.

2. Aesthetics of the 1920s (Art Déco)

After 1918, the spirit of the times changed fundamentally. Nature was replaced by the machine; design became urban and technocratic.

- **Geometric discipline:** Rectangles, triangles, stepped forms and the meander (Greek key) dominate. Geometry is not a filler but an architectural directive.
- **Industrial rhythm:** Patterns such as the meander or repeating ovals are designed for industrial mass production (stamping). They appear cool, predictable and precise.
- **Veneer design:** Walnut is used in large surfaces, not as a base for carvings. A simple black-and-white contrast border (stringing) underscores the geometry of the wood.

Accordion Construction

- **Fittings:** Use of cool nickel silver or nickel. Fittings now mark the corners rather than “decorating” them, which emphasises the construction of the instrument.

3. The Collector’s Trap: Historicist Geometry

A common mistake is confusing pre-war geometry with Art Déco.

- **Pre-war period (geometric historicism):** Here the *horror vacui* (fear of empty space) prevails. The surface is completely covered with small, often overloaded geometric inlays. Every element is individually fitted. This is a demonstration of craftsmanlike patience, not industrial efficiency.
- **Distinction:** Ask yourself: does the pattern look like a densely woven, hand-crafted carpet (pre-war) or like a clear, rhythmic skeleton (1920s)?

Conclusion

The decoration of an accordion is a mirror of its era. While Art Nouveau wanted nature to grow into the instrument, Art Déco sought to define the instrument as a technical, precise object of modernity. With floral motifs, focus not on the “what” (e.g. a lily) but on the “how” (is the line organically fluid or statically geometric?).

Afterword

The reed is one of the oldest and least understood sound-generating principles in musical instruments. It is mechanically simple -- a thin metal strip vibrating through a slot -- yet acoustically complex: the interaction between reed, gap, chamber and bellows produces behaviour that defies simple models.

This document collection has attempted to work through the physical foundations of this interplay -- not with a claim to completeness, but with the aim of making the essential mechanisms visible. From the flow analysis at the reed gap through the impedance coupling of the chamber to the audibility of individual overtone modes, a common thread runs through: accordion acoustics can be described with the same tools proven in electronics and communications for decades -- resonance, impedance, quality factor, coupling.

Some things in these documents are established, some are models and estimates. I have indicated the boundary between the two to the best of my knowledge. Where simplified models reach their limits, this is noted. The accompanying Python scripts allow every calculation to be verified independently and checked against your own measurements.

Instrument making lives in the workshop, not at the desk. No document replaces listening, testing and correcting on the actual instrument. But an understanding of the physics can help to ask the right questions -- and translate experience more quickly into good results.

I hope this collection proves useful to other instrument makers, repairers and interested readers. Anyone who finds errors, knows better models, or wishes to contribute their own measurements is warmly invited to contact me. The documents and calculation scripts are available on GitHub and will be updated as needed.